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Citation (Version of Record):
Guo, L., Sang, B., Tu, J., & Wang, Y. (2024). Cross-cryptocurrency return predictability. *Journal of Economic Dynamics and Control*, *163*, Article 104863. https://doi.org/10.1016/j.jedc.2024.104863

Version:
Peer reviewed version

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Link to published version (if available):
10.1016/j.jedc.2024.104863

Link to publication record on the Bristol Research Portal:
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Cross-cryptocurrency Return Predictability*

Li Guo†

School of Economics, Fudan University

Shanghai Institute of International Finance and Economics

Bo Sangξ

School of Business, University of Bristol

Jun Tuτ

SJTU-BOC Institute of Technology and Finance, Antai College of Economics and Management,

Shanghai Jiao Tong University

Lee Kong Chian School of Business, Singapore Management University

Yu Wang‡

School of Finance, Shanghai University of Finance and Economics

* We are grateful to helpful comments from two anonymous reviewers, the associated editors and the editor, and Weikai Li, Hong Liu, Yukun Liu, David Rapach, Guofu Zhou, Yingzi Zhu, along with seminar participants at the Singapore Management University, Fudan University, and Zhejiang University. Li Guo thanks the financial support by the National Natural Science Foundation of China (#72003040) and the Chenguang Program of Shanghai Education Development Foundation and Shanghai Municipal Education Commission for his work on the project. Tu acknowledges financial support from the Singapore Ministry of Education (MOE) Academic Research Fund (AcRF) Tier 1 (Fund no.: C207 / MSS20B008) grant.

† School of Economics, Fudan University. 600 Guoquan Road, Shanghai 200433. Email: guo_li@fudan.edu.cn.

ξ School of Business, University of Bristol. 15-19 Tyndall's Park Rd, Bristol, United Kingdom BS8 1PQ. Email: bo.sang @bristol.ac.uk.

τ Corresponding author. SJTU-BOC Institute of Technology and Finance, Antai College of Economics and Management, Shanghai Jiao Tong University. 1954 Huashan Road, Shanghai, China, 200030. Phone: (+86) 21-52301159. Email:tujun2022@sjtu.edu.cn. Lee Kong Chian School of Business, Singapore Management University. 50 Stamford Rd, Singapore 178899. Email: tujun@smu.edu.sg. Phone: (+65) 68280764. Fax: (+65) 68280427.

‡ School of Finance, Shanghai University of Finance and Economics. 777 Guoding Road, Shanghai 200433. Email: wang_yu@163.sufe.edu.cn.

Cross-cryptocurrency Return Predictability

Abstract

Using data from *Binance*, we find strong evidence of cross-cryptocurrency return predictability. The lagged returns of other cryptocurrencies serve as significant predictors of focal cryptocurrencies. The results are robust across various methods, including the adaptive LASSO and principal component analysis. Furthermore, a long-short portfolio formed on the past returns of cryptocurrencies can generate a sizable return out-of-sample after accounting for transaction costs. Overall, our findings corroborate cross-cryptocurrency return predictability and are consistent with the spillover effect mechanism, where common shocks among cryptocurrencies coupled with the limited attention of investors lead to slow information diffusion across coins.

JEL classifications: G10, G11, G14, G40

Keywords: Cryptocurrency; return predictability; information spillover; adaptive LASSO

1. Introduction

Ever since the launch of Bitcoin, the cryptocurrency landscape has seen a significant surge in investments, amassing trillions of funds over the past decade. [1] However, valuing cryptocurrencies poses a challenge when compared to traditional assets. Traditional assets often come with a plethora of publicly available predictive information variables, such as analyst coverage and accounting data. Cryptocurrencies, in stark contrast, lack many of these tangible metrics. This makes them particularly elusive in terms of valuation (e.g., Detzel, Liu, Strauss, Zhou, and Zhu, 2020).

Combined with the 24/7 trading nature of cryptocurrencies, it is not surprising that cryptocurrencies exhibit more volatile price movements and are often subject to intense speculation (e.g., Brauneis and Mestel, 2018; Sockin and Xiong, 2021). The difficulties in pining down the valuation and returns of cryptocurrencies tend to be further intensified by factors like the digital mining processes, investor attention limits, and the pervasive influence of algorithmic and high-frequency trading.

With those challenges and difficulties, there have been attempts to demystify cryptocurrency returns – using information from initial coin offerings (ICO) and historical coin returns. Nevertheless, these endeavors typically revolve around individual cryptocurrency data and characteristics.[2] [3] In other words, these studies are based on each individual cryptocurrency's own information and characteristics. The question that then arises, which forms the crux of our study,

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[1] More than 6,000 cryptocurrencies are actively traded nowadays, and their overall market capitalization is above $1.5 trillion U.S. dollars as of February 2021, which is almost 3% of the total U.S. stock market capitalization, according to *CoinMarketCap*.

[2] See, for example, Hou, Li, Liao, and Zhang (2019); Lee, Li, and Shin (2021); Detzel, Liu, Strauss, Zhou, and Zhu (2020); Liu, Sheng, and Wang (2021); Liu and Tsyvinski (2021); Liu, Tsyvinski, and Wu (2021).

[3] The terms cryptocurrencies, crypto, and coins are used interchangeably in this paper.

is whether the information (e.g., historical returns) from one cryptocurrency can forecast returns of another, i.e., cross-cryptocurrency return predictability.

Unlike traditional assets such as equities or fiat currencies, cryptocurrencies operate within a unique decentralization market structure, primarily rooted in blockchain technology. Such shared technological underpinnings suggest that innovations across cryptocurrencies are likely to experience common shocks and knowledge spillovers (Jaffe, Trajtenberg, and Henderson, 1993).[4] This can significantly impact cryptocurrency returns, leading to lead-lag effects among returns (Lee, Sun, Wang, and Zhang, 2019). For instance, Bitcoin (BTC), a leading cryptocurrency, often exhibits more immediate reactions to common shocks than many of its counterparts. Smaller coins often showcase delayed reactions, potentially due to limited attention or constrained information-processing capabilities of their investor base (e.g., Hong, Torous, and Valkanov, 2007; Merton, 1987; Shahrur, Becker, and Rosenfeld, 2010).

With this backdrop, our study hypothesizes the prevalence of cross-cryptocurrency return predictability. To substantiate this hypothesis, we have gathered minute-frequency pricing data of cryptocurrencies from a leading exchange, *Binance*. To avoid survival bias, we select the top 30 coins with the highest trading volume on *Binance* at the time that we started the project. Our sample spans from 25 March 2019 to 30 April 2021 in the 24/7 cryptocurrency spot market. As of the day we started the project (9 May 2020), the dollar trading volume of the 30 coins accounts for more than 86% of the total coin market dollar trading volume, ensuring that our analysis is representative of the cryptocurrency market.

[4] For example, a lot of cryptocurrencies use the same consensus algorithm as the Bitcoin (Proof of Work) and one common feature of such cryptocurrency platforms is that they involve a group of miners to verify or record transactions on the blockchain (Sockin and Xiong, 2021). Other papers that shed light on the underlying technology of cryptocurrencies include Cong and He (2019) and Cong, He, and Li (2021).

Univariate ordinary least squares (OLS) regressions show that the returns of all 30 cryptocurrencies are significantly influenced by the lagged returns of other cryptocurrencies. Moreover, most coins exhibit positive coefficients on the lagged returns of other coins, consistent with our argument about the lead-lag effect among returns of cryptocurrencies. To reduce statistical concerns associated with correlated regressors, we choose the following two empirical frameworks: 1) pooled estimates with bias-corrected wild bootstrapped confidence intervals (e.g., Rapach, Strauss, and Zhou, 2013); and 2) the adaptive version of least absolute shrinkage and selection operator (LASSO) in the predictor estimation (Tibshirani, 1996; Zou, 2006). In terms of the pooled estimation, returns of about half of the sample coins can be positively and significantly predicted by the pooled lagged returns of other coins. A one standard deviation increase in the pooled lagged return of other coins will increase an individual coin's return by 0.40 bps – 4.82 bps on a minute basis. The adaptive LASSO identifies lagged returns for at least six other coins as significant return predictors for any one of the 30 coins. Particularly, BTC has been selected by the algorithm as a positive and significant return predictor for all other coins except for stablecoins. Other top return predictors are Binance coin (BNB) and Tron (TRX), which also positively predict returns for all other coins excluding stablecoins. Overall, these results strongly corroborate cross-cryptocurrency return predictability.

In addition, we test the robustness of our results by utilizing the principal component analysis (PCA) approach. We extract three main principal components from the 30 cryptocurrency returns on a rolling basis. All cryptocurrencies in our sample, except for stablecoins, have positive and relatively uniform loadings on the first principal component, suggesting that the 30 cryptocurrency returns can be affected by some common shocks. The varying loadings on the two remaining principal components indicate that these two principal components may capture certain

complex coin interdependencies in the cryptocurrency market. According to the distribution of the return variance among the three main principal components, the first principal component represents 36.80% of the total variance, while the other two main principal components represent just 8.26%. These ratios indicate that the common shocks may play a much stronger role in cross-cryptocurrency return predictability than the complex coin interdependencies.

To measure the economic significance of the forecasting power, we implement multiple trading strategies and evaluate the out-of-sample performances of corresponding portfolios. Specifically, we sort cryptocurrencies into equal-weighted quintile portfolios and construct zero-investment long-short portfolios, based on the out-of-sample cryptocurrency return forecasts (minute-level) via the adaptive LASSO, PCA, and the univariate (BTC) regression, respectively. Portfolio returns increase monotonically from the lowest quintile to the highest quintile in all three prediction models, and the minute-level return spread for these three long-short portfolios are 3.34 bps, 1.85 bps, and 1.54 bps, respectively. The t-statistics of the returns are larger than three and thus meet the requirement of the multi-hypothesis threshold posed in Harvey (2017), alleviating the p-hacking problem.

We demonstrate that the returns of these long-short portfolios cannot be explained by exposures to various factors in the cryptocurrency market, including market risk, size, and alternative specifications of momentum. Additionally, we employ size, Google search volume, and distraction to proxy for investor attention. Our sub-sample tests reveal larger alphas for coins that attract lower investor attention. We also dismiss the alternative explanation that liquidity drives cross-cryptocurrency predictability, using both Amihud's (2002) illiquidity and abnormal trading volume as proxies for coin liquidity. Moreover, our primary results remain consistent across different market conditions, indicating that neither market booms nor market uncertainties

influence the cross-crypto return predictability observed in our study. A series of robustness tests further supports the causal relationship inherent in the spillover channel.

We also investigate nine events (e.g., PayPal's adoption of cryptocurrencies as a payment tool) that have had broad impact on the cryptocurrency market and examine the corresponding long-short portfolio alphas during the days with those nine major events (event days). Compared with the average alpha for the other days outside those event days with no events or some minor events sporadically (labelled as normal days hereafter), the alphas on the event days are significantly higher than on the normal days. This is because those market-wide events should affect various coins as common shocks, thereby resulting in abnormally larger common shocks to various coins and a consequential higher level of cross-cryptocurrency return predictability on the event days than on the normal days. Overall, these results are consistent with the spillover-effect mechanism, where common economic linkages among coins coupled with the limited attention of investors cause slow information diffusion.

Moreover, by investigating the out-of-sample performance using the adaptive LASSO strategy in the spot market, we find that the returns are economically and statistically significant. Since short selling is difficult in practice for the spot market, we also examine the portfolio performance in the cryptocurrency futures markets and find that the long-short portfolio still yields sizable gains with statistical significance after accounting for trading costs. For example, a long-short portfolio based on quintile sorting using a half-day training window exhibits a return of 0.34 bps for regular takers (with trading costs of 4 bps) with 10-minute rebalancing, which is roughly 3% per week and comparable to the performances of trading strategies of cryptocurrencies (e.g., Liu, Tsyvinski, and Wu, 2021). The portfolios perform even better when we employ the rebalancing frequency as five minutes.

One of the primary contributions of our study is demonstrating the prevalence of cross-cryptocurrency return predictability. This augments the expanding literature on asset pricing studies within the cryptocurrency market. For instance, Lee, Li, and Shin (2021) found that analyst ratings for initial coin offerings (ICOs) positively predict long-term cryptocurrency returns. Liu, Tsyvinski, and Wu (2021) documented a robust time-series momentum effect and highlighted that investor attention is a potent predictor of future cryptocurrency returns. Other studies indicate that factors such as market size and volatility aptly capture expected cryptocurrency returns (e.g., Li and Yi, 2019; Liu and Tsyvinski, 2021). Yet, the realm of cross-cryptocurrency return predictability remains largely uncharted.

While numerous studies have delved into the interdependencies among leading cryptocurrencies and documented connectedness across coins (e.g., Fry and Cheah, 2016; Ciaian, Rajcaniova, and Kancs, 2018; Corbet et al., 2018; Antonakakis, Chatziantoniou, and Gabauer, 2019; Chaim and Laurini, 2019; Palamalai and Maity, 2019; Ji, Bouri, Lau, and Roubaud, 2019; Shams, 2020), they predominantly concentrate on contemporaneous relationships or do not explicitly tackle return predictability. Our paper distinguishes itself from these studies and contributes uniquely by showing that the lagged returns of certain cryptocurrencies can be significant predictors for the returns of other cryptocurrencies.

Additionally, our work offers insights into the information spillover effect within the cryptocurrency market. A plethora of earlier studies have documented the spillover effect in stock and bond markets.[5] They highlight that common economic linkages between firms or industries,

[5] See, for example, Lo and MacKinlay (1990); Brennan, Jegadeesh, and Swaminathan (1993); Badrinath, Kale, and Noe (1995); Chordia and Swaminathan (2000); Cohen and Frazzini (2008); Menzly and Ozbas (2010); Shahrur, Becker, and Rosenfeld (2010); Rapach, Strauss, Tu, and Zhou (2019); Lee, Sun, Wang, and Zhang (2019); Ali and Hirshleifer (2020).

combined with factors such as limited attention or constrained information-processing capabilities, lead investors to underreact to shocks in interconnected firms or sectors. Regarding the cryptocurrency market, some researchers have explored spillover effects between cryptocurrencies and other elements, like traditional currencies (Al-Shboul, Assaf, and Mokni, 2022), economic uncertainty (Wang, Xie, Wen, and Zhao, 2019), and energy and technology companies (Symitsi and Chalvatzis, 2018). To the best of our knowledge, our research is pioneering in examining cross-cryptocurrency return predictability rooted in the information spillover effect using minute-level data. Katsiampa, Corbet, and Lucey (2019) and Omane-Adjepong and Alagidede (2019) document volatility spillover effects using data of a few major cryptocurrencies. Examining return connectedness among cryptocurrencies, Moratis (2021) conducts a time series in-sample analysis on the risk spillover. However, the volatility or risk spillover explanation differs from that of the information spillover: the former is about spillover shocks that could affect returns in either direction while the latter mainly result in a positive lead-lag effect. Sifat, Mohamad, and Shariff (2019) use various statistical methods to examine the lead-lag relationship between Bitcoin and Ethereum and find no clear pattern. Katsiampa, Corbet, and Lucey (2019), as well as Omane-Adjepong and Alagidede (2019), have documented volatility spillover effects with data from a select group of prominent cryptocurrencies. Investigating return connectedness among cryptocurrencies, Moratis (2021) performed a time series in-sample analysis focusing on risk spillover. However, the rationale for volatility or risk spillover differs from information spillover: the former pertains to spillover shocks that can influence returns in any direction, while the latter predominantly results in a positive lead-lag effect. Sifat, Mohamad, and Shariff (2019) employed various statistical methods to study the lead-lag relationship between Bitcoin and Ethereum, finding no consistent pattern.

Building on the insightful work of Jia, Wu, Yan, and Liu (2023), which documents a significant negative lead-lag effect known as the "seesaw effect" in cross-cryptocurrency returns, our study introduces a complementary narrative focusing on a positive information spillover effect. This divergence in findings could be attributed to differences in the sample periods and datasets employed. Jia et al. (2023) analyzed data from *Bitfinex* and *Huobi*, spanning 1 January 2018 to 30 July 2021, a timeframe that, given the broader context of their findings, likely reflects a market environment with a predominant presence of retail trading behaviors. In contrast, our analysis utilizes data from *Binance*, covering 25 March 2019 to 30 April 2021, which has seen an increasing enthusiasm of institutional investors.

The choice of *Binance* for our study is informed by its significant global presence and appeal to institutional investors, factors that could influence the dynamics of information spillover across cryptocurrencies. The growing participation of institutional investors, as documented by Huang, Lin, and Wang (2022), marks a shift in market dynamics, potentially moving away from the dynamics observed in earlier periods or different exchange environments such as those analyzed by Jia et al. (2023).

Our research sheds light on how different market conditions and investor behaviors, particularly the potential shift towards a greater institutional presence in later periods and on platforms like *Binance*, can influence the observed lead-lag effects. This perspective adds to the existing literature, including the work of Jia et al. (2023), by highlighting the role of investor demographics in shaping the nature of information spillover across cryptocurrencies.

Moreover, we employ a machine learning approach to effectively amalgamate various predictors, forming trading strategies that probe the investment value of the information spillover effect. Our efforts align with the burgeoning literature that integrates machine learning techniques

into empirical asset pricing (e.g., Kozak, Nagel, and Santosh, 2020; Freyberger, Neuhierl, and Weber, 2020; Gu, Kelly, and Xiu, 2020; Da, Nagel, and Xiu, 2022; Chen, Hu, and Yang, 2022). Crucially, by illuminating the information transmission mechanism within the cryptocurrency market, our paper lends support to the gradual information diffusion theory as an explanation for the lead-lag effect (e.g., Hong and Stein, 1999; Hong, Lim, and Stein, 2000; Hou, 2007).

Our research also complements the extant literature on high-frequency trading within the cryptocurrency domain. In recent times, the pronounced volatility of the cryptocurrency market has drawn a multitude of high-frequency traders, who deploy algorithms for swift, large-scale trades. Various academic endeavors have dissected intra-day cryptocurrency trading data, unveiling insights related to high-frequency trading and return patterns (e.g., Makarov and Schoar, 2020; Liu, Marsh, Mazza, and Petitjean, 2021). Our study broadens this body of work, adopting a holistic approach that analyzes the top 30 cryptocurrencies – comprising 86% of the entire cryptocurrency market by dollar trading volume – and delves into cross-cryptocurrency return predictability from a high-frequency vantage point.

The rest of this paper is organized as follows. Section II introduces the data. In section III, we present the empirical findings, including various robustness tests. Section IV contains concluding remarks.

2. Data and summary statistics

We obtain pricing information and trading activities of 30 cryptocurrencies from *Binance*, a leading global cryptocurrency exchange.[6] To avoid the survival bias, we select the coins based

[6] *Binance* has been ranked the top crypto exchange with the highest trading volume and the largest number of weekly visits. Source: https://coinmarketcap.com/rankings/exchanges/.

on their large trading volume on *Binance* at the time when we started the project, i.e., 9 May 2020. The sample period spans from 25 March 2019 to 30 April 2021, and the data frequency is on a minute basis in the 24/7 cryptocurrency spot market, to ensure that the machine learning method avails of an adequate amount of data. For the cryptocurrency futures market, the sample covers from 29 July 2020 to 30 April 2021. The starting date is the first day that all 30 coins were traded on *Binance*. We further standardize coin returns – with means equal to zero and standard deviations equal to one – to make it comparable across different coins. All prices are dominated in Tether (USDT).[7] The trading volume of the top 30 coins included in the sample amounts to $1.3 billion U.S. dollars, representing more than 86% of the total cryptocurrency market dollar trading volume (see Figure A.1 in the Appendix). Therefore, our analysis of the 30 top cryptocurrencies is indeed representative of the overall cryptocurrency market.

Table I presents the summary statistics of the 30 cryptocurrencies in our sample. It shows that all cryptocurrencies except for stablecoins display a time-series average of the minute-level return larger than 0.01 bps, which is equivalent to an hourly return of 0.60 bps. The average standard deviation of sample coins' minute-level returns is as high as 0.19%, indicating the highly volatile nature of the cryptocurrency market. This rampant price movement of cryptocurrencies is consistent with the literature (e.g., Brauneis and Mestel, 2018; Sockin and Xiong, 2021). In contrast, TUSD, USDC, and PAX, which are the three USD-pegged stablecoins[8] included in our sample, have a zero mean of returns and exhibit the lowest standard deviations. Such zero mean returns and low volatility features are not surprising as the values of these stablecoins are pegged to fiat currency (USD) rather than linked to blockchain protocol, platform utility, or network

[7] Tether is a popular unit symbol used in exchanges, and it is 1:1 pegged to the USD.
[8] According to Chohan (2021), stablecoins have emerged in the recent past with an approach to mitigate the high volatility of cryptos through a sustained peg with traditional instruments such as USD or a basket of currencies.

activities (e.g., Chohan, 2021; Hoang and Baur, 2021). Therefore, they may not share common information with other cryptocurrencies in our sample.

<To insert Table I here.>

3. Empirical results

3.1. Baseline models

We first run a univariate ordinary least squares (OLS) regression model for each sample coin as follows:

$$r_{i,t} = \hat{\alpha}_{i,t} + \hat{\beta} r_{j,t-1} + \epsilon_{i,t} \quad (1)$$

where $r_{i,t}$ is the return of cryptocurrency i at time t (minute-level), $r_{j,t-1}$ is the returns of all cryptocurrencies in the sample at time t-1, $\epsilon_{i,t}$ is the error term, and i, j = 1, …, 30. When j = i, Equation (1) explores the relationship between the coin's own lagged return and its current return. Results are reported in Table A.I. The main finding is that returns of all 30 sample cryptocurrencies are largely influenced by the lagged returns of other cryptocurrencies. Specifically, returns of 24 out of 30 coins can be significantly predicted using the lagged returns of all other coins.[9] Moreover, most coefficients are positive, which is consistent with the lead-lag spillover effect documented in the stock market (e.g., Lee, Sun, Wang, and Zhang, 2019). Overall, in the context of the cryptocurrency market, given the limited attention or constrained information-processing capabilities of investors (Subramaniam and Chakraborty, 2020), common information is not homogeneously incorporated into individual coins, so the lagged returns of other coins may contain valuable information that is incorporated into the prices of the focal coins with a delay,

[9] As for the remaining six coins, the average number of other coins that significantly predict their returns is as high as 26 out of 29.

thereby engendering positive cross-cryptocurrency return predictability. As the leading coins in the cryptocurrency market, BTC and ETH may respond to new information more promptly than many other coins. As a result, the positive lead-lag effect is weaker for them and their loadings on the lagged returns of the other coins are not always positive. Such occasional negative loadings may capture some complex coin interdependencies in the cryptocurrency market, which may be dominated by the positive lead-lag effect for smaller coins but significantly affect large coins where the positive lead-lag effect is weak.

Another finding is that all cryptocurrencies except for ONT and ZEC have negative coefficients on their own lagged returns. This intraday return reversal pattern is also documented in Petukhina, Reule, and Härdle (2021). To some extent, for each focal coin, if common shock affects its own price gradually with a delay, the prediction sign of its own lagged return should be positive as well, same as the sign of lagged returns of other coins. We provide some explanation on this as follows. Given Bitcoin's substantial share in the cryptocurrency market, it should attract a lot of attention from investors. Thus, a positive common shock at time t-1 is likely to cause BTC to have an increase in return at time t-1 without much delay. However, for a smaller cryptocurrency, such as ADA, its return at t-1 may not be affected much by this common shock due to investors' limited attention on small cryptocurrencies. Instead, its return may decrease at t-1 if it faces a large downward price pressure.[10] Such price pressure impact can be treated as independent from that of the common information.[11] Meanwhile, BTC is not likely to face a large downward price pressure same as ADA at t-1 as evidenced by the low correlations among the

[10] See, for example, Kraus and Stoll (1972), Duffie (2010), and Hendershott and Menkveld (2014).
[11] Specifically, according to Campbell, Grossman, and Wang (1993), the price pressure is caused by selling or buying pressure from the "noninformational" traders, who have not processed the common information reflected in the lagged returns of peer coins, indicating that the common information shocks and the price pressure shocks tend to be uncorrelated.

contemporaneous returns of the cryptocurrencies[12] and is still able to have a positive return increase driven by the positive common shock. In the following period t, on the one hand, the focal coin, ADA, is likely to experience a return reversal when the downward price pressure starts to taper off from its own lagged return.[13] On the other hand, when the positive common shock starts to be incorporated into the price of ADA, the return of ADA tends to increase. As such common shock also drives the lagged return of BTC at time t-1, we observe the positive cross-crypto return predictability. Meanwhile, given that the common shock effect and the ADA-specific price pressure tend to be uncorrelated (see footnote 12), the return reversal of ADA on its own lagged return due to the ADA-specific price pressure is not likely to be affected by the common shock. Overall, Table A.I suggests that cross-cryptocurrency return predictability is indeed prevalent, and there is also a wide-spread return reversal pattern for the lagged returns of focal coins[14].

Since BTC has been the largest coin in terms of market capitalization and an essential cryptocurrency market driver, it is expected to react to common information shocks in the cryptocurrency market faster than other cryptocurrencies. According to the gradual information diffusion theory, coins with a larger investor base (proxied by market capitalization) are expected to lead returns of smaller coins, in the spirit of Hou (2007). Therefore, we take Bitcoin's influence on the returns of other cryptocurrencies into account by including the lagged return of BTC as a regressor into the univariate regression. We also include its own lagged return as the control variable, as follows:

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[12] The average correlation is 0.30 among the returns of the top 30 coins, with the first and third quartiles being 0.15 and 0.43, respectively.

[13] Temporary price pressure and the subsequent return reversal in a short horizon are documented in both equity and cryptocurrency literature (e.g., Heston, Korajczyk, and Sadka, 2010; Petukhina, Reule, and Härdle, 2021).

[14] The three stablecoins (PAX, TUSD, USDC) have negative loadings on the lagged returns of other coins, probably because their values are pegged to fiat currency rather than determined by the blockchain protocol, platform utility or network activities. Hence, they do not share common information with other coins in our sample.

$$r_{i,t} = \hat{\alpha}_{i,t} + \hat{\beta}_{i,BTC}\, r_{BTC,t-1} + \hat{\beta}_{i,i}\, r_{i,t-1} + \epsilon_{i,t} \quad (2)$$

where i = 1, …, 29 (excluding BTC).

Table II presents the estimated coefficients of the lagged BTC return in Equation (2). The main finding here is the information spillover effect of BTC on other coins: the coefficients $\hat{\beta}_{i,BTC}$ are all statistically significant at the 1% level. For all cryptocurrencies, except for the three stablecoins (PAX, TUSD, USDC), the lagged BTC return has positive predictability for their returns. As all return variables are standardized, a one standard deviation increase in the BTC lagged return will increase the return of other coins by 1.92 bps, which is equivalent to 1.15% on an hourly basis. By contrast, the lagged BTC return has negative predictability for the returns of the three stablecoins. Overall, Table II indicates that BTC wields a strong information spillover effect upon the returns of other cryptocurrencies, with both statistical and economic significance.

<To insert Table II here.>

Jia et al. (2023), however, document a significant negative lead-lag effect in cross-cryptocurrency returns using a 5-minute frequency. To reconcile our baseline findings with those of Jia et al. (2023), we use 5-minute cryptocurrency returns to repeat our above analysis. The results, shown in Table A.II, remain consistent with our finding in Table II, where the lagged BTC return positively predict other coins' returns except for stablecoins. Therefore, changes in data frequency alone do not fully explain the discrepancies between our findings and those of Jia et al. (2023). One possible reason for these differences could be the distinct crypto exchange data and sample periods used in each study.[15] To delve deeper, we applied the methodology of Jia et al.

[15] While Jia et al. utilized data from *Bitfinex* and *Huobi*, our research is based on data from *Binance*. Moreover, our sample period (03-25-2019 to 04-30-2021) does not cover the earlier timeframe analyzed by Jia et al. (01-01-2018 to 07-30-2021), despite their extended sample period in the published 2023 version that includes our study period.

(2023) to obtain data from *Bitfinex* for the same 22 coins during our sample period. The results, shown in Table A.III, reveal that lagged BTC returns positively and significantly predict the 5-minute returns of 20 out of 22 coins. This, coupled with the negative lead-lag effect observed by Jia et al. (2023), suggests a temporal shift in the lead-lag effect.

The initial period studied by Jia et al., which falls outside our sample, likely featured considerable naive retail trading, leading to a "fight-to-hot-coins" and a negative lead-lag effect. Conversely, our later sample period shows a positive lead-lag effect, potentially due to an influx of institutional investors, as highlighted by Huang, Lin, and Wang (2022),[16] fostering greater information spillovers across coins. These sophisticated institutional investors, through their trades, likely contribute to information spillovers across coins, in contrast to the "fight-to-hot-coins" effect predominantly driven by naïve retail trading. In line with the gradual information diffusion theory (Hou, 2007), we would expect large coins to lead the returns of smaller coins. This lead-lag effect is further accentuated in our sample period, likely due to the tendency of institutional investors to trade larger coins rather than smaller ones.[17][18]

We provide further evidence for the cross-cryptocurrency return predictability, by running multivariate OLS regressions with lagged returns of all cryptocurrencies as the independent variables. Specifically, we run the following regression model:

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[16] Additionally, the *Binance* report in 2023 observes that over a third (35.6%) of the institutional investors increased their allocation over the year. Source: https://research.binance.com/static/pdf/institutional-crypto-outlook-survey-2023.pdf.

[17] The Cointelegraph Research survey in 2022 shows that 94% of the institutional investors invest in Bitcoin, and 75% of them also invest in Ethereum. Around 13% of them invest in Solana or Litecoin. They do not seem to invest much in small coins. Source: https://cointelegraph.com/news/most-institutional-investors-hold-or-plan-to-buy-cryptocurrencies-survey.

[18] In our unreported tables, we divide our dataset based on the median date of our sample period into halves and repeat our baseline analysis same as in Table A.I and Table II. We indeed find a significantly stronger positive predictability in the more recent sub-sample.

$$r_{i,t+1} = \alpha_i + \sum_{j=1}^{N} \beta_{i,j} r_{j,t} + \epsilon_{i,t+1} \quad (3)$$

where $r_{i,t+1}$ is the return of cryptocurrency i at minute t+1 (where i = 1, …, 30), N is the total number of cryptocurrencies, and $\epsilon_{i,t+1}$ is the error term. Because a plethora of correlated regressors incur issues such as overfitting and imprecise parameter estimates, we therefore improve the multi-factor OLS regression model by using pooled estimates to increase estimation efficiency, following the approach of Rapach, Strauss, and Zhou (2013). Table III reports pooled OLS coefficient estimates with bias-corrected wild bootstrapped 90% confidence intervals (shown in brackets). The results further corroborate our previous findings: for about half of the sample coins (14 out of 30), their returns can be significantly and positively predicted using the pooled lagged returns of other coins. The economic magnitude is large as well, since a one standard deviation increase in the pooled lagged return of other coins will cause the returns of these 14 coins to increase by 0.40 bps – 4.82 bps on a minute basis, which is equivalent to 24 bps – 289 bps hourly. Again, the large and positive coefficients indicate that the common information of the cryptocurrency market is not homogeneously incorporated into individual coins, suggesting the presence of the information spillover effect in the cryptocurrency market.

< To insert Table III here. >

3.2. Adaptive LASSO method

The aforementioned measure, using pooling estimates and bias-corrected wild bootstrapping, does alleviate, but does not solve, the drawbacks of multivariate OLS regression with an excessive number of regressors. To further circumvent such statistical issues, we follow the statistics literature by employing the adaptive version of least absolute shrinkage and selection operator (LASSO) in the predictor estimation (e.g., Tibshirani, 1996; Zou, 2006; Rapach, Strauss, Tu, and Zhou, 2019; Chinco, Clark-Joseph, and Ye, 2019; Chen, Hu, and Yang, 2022). In short, LASSO

performs both shrinkage and variable selection[19] while the adaptive LASSO proposed by Zou (2006) further improves the estimation in terms of better variable selection accuracy and consistency. The adaptive weights in the adaptive LASSO can also reduce the bias associated with LASSO especially for a high-dimensional dataset. We thus use the adaptive LASSO in this paper[20]. The adaptive LASSO estimates for the general predictive regression model in Equation (3) are:

$$\widehat{\boldsymbol{\beta}}_{\boldsymbol{i}}^{*} = argmin\left\| r_{i,t+1} - \sum_{j=1}^{N} \beta_{i,j}\tilde{r}_{j,t} \right\|^{2} + \lambda_i \sum_{j=1}^{N} \widehat{w}_{i,j} \left|\beta_{i,j}\right| \quad (4)$$

where $r_{i,t+1}$ is the return of cryptocurrency i at minute t+1 (i = 1, …, 30), N is the total number of cryptocurrencies, i.e., 30, $\tilde{r}_{j,t}$ is the standardized return for cryptocurrency j, $\widehat{\boldsymbol{\beta}}_{\boldsymbol{i}}^{*} = (\hat{\beta}_{i,1}^{*}, \ldots, \hat{\beta}_{i,N}^{*})'$ is the N-vector of adaptive LASSO estimates, λ_i is a non-negative regularization parameter, and $\widehat{w}_{i,j}$ is the weight vector of $\left|\beta_{i,j}\right|$ for j =1,…, N. In particular, $\lambda_i \sum_{j=1}^{N} \widehat{w}_{i,j} \left|\beta_{i,j}\right|$ is the ℓ_1 penalty that allows the coefficient estimates to shrink continuously towards zero as λ_i increases. We also follow Zou (2006) to define the weight vector $\widehat{w}_{i,j}$ as:

$$\widehat{w}_{i,j} = 1/\left|\hat{\beta}_{i,j}\right|^{\gamma_i} \quad (5)$$

where $\hat{\beta}_{i,j}$ is the OLS estimate of $\beta_{i,j}$ and γ_i>0.

Table A.IV presents the adaptive LASSO estimates of the general predictive model for all sample cryptocurrencies. To conserve space, we use bold to denote statistically significant coefficients with bootstrapped 90% confidence intervals or higher. Consistent with the results of univariate OLS regressions, the adaptive LASSO estimates uncover both the information spillover

[19] LASSO is widely used in finance literature. For example, Chinco, Clark-Joseph, and Ye (2019) use LASSO to identify stock return predictors and find it increases both out-of-sample performance and Sharpe ratios of corresponding trading strategies. Recent theoretical papers in machine learning such as Chen, Hu, and Yang (2022) and Da, Nagel, and Xiu (2022) also lend support to the use of LASSO in predicting asset returns.

[20] We also report the regression results using the LASSO method in Table A.VI and the results remain robust.

effect and the return reversal pattern in the cryptocurrency market. Indeed, after controlling for lagged return, spillover effects remain strong across all the regression models. Based on machine learning, the adaptive LASSO technique helps select variables in predictive regressions. According to Table A.IV, it identifies at least six other coins' lagged returns as significant return predictors for each sample coin. The average number of other coins selected as informative return predictors for each focal coin is 23, which represents 77% of the sample size. Hence, the results of the adaptive LASSO approach reinforce the findings in OLS regressions that the information spillover effect is indeed prevalent in the cryptocurrency market. Consistent with stock spillover literature (e.g., Cohen and Frazzini, 2008; Menzly and Ozbas, 2010; Lee, Sun, Wang, and Zhang, 2019), we show that in the cryptocurrency market, individual cryptocurrencies do not homogeneously incorporate the common information. Hence, the other coins' lagged returns contain valuable information that is slowly incorporated into the prices of the focal coins, thereby causing cross-cryptocurrency return predictability.

Compared to the coefficients generated from the univariate OLS regression, the number of significant predictors in Table A.IV has been largely reduced for many coins in our sample. The variable-dropping phenomenon may be attributed to the improved estimation accuracy by using the adaptive LASSO technique. Also, some coefficients exhibit negative signs in Table A.IV. This could stem from the high correlations among different coins. As shown in Table A.V, each coin exhibits positive and nonnegligible correlation with other coins, except for stable coins with low correlations. For each coin, the average correlation (excluding the three stable coins: PAX, TUSD, USDC) with other coins is as high as 0.44. The high correlations may incur potential multicollinearity issue that could result in negative coefficients for some coins. However, these coins are mostly smaller coins in our sample and tend to respond to common shocks with a delay.

While those larger coins tend to still have positive coefficients.

Among the return predictors, Bitcoin is selected by the adaptive LASSO technique to positively predict returns for all sample coins except for stablecoins. Its predictability is both statistically and economically significant. Specifically, a one standard deviation increase in BTC lagged returns will increase the returns of other coins (excluding stablecoins, which, due to their special nature of being pegged to fiat currency, exhibit negative coefficients in this case) by 1.25 bps on average, which is equivalent to 75 bps on an hourly basis. This information spillover effect of BTC on other coins is also documented in Table II. A possible explanation is that BTC, as the first widely circulated cryptocurrency and the largest in terms of market capitalization, serves as an essential cryptocurrency market driver and tends to respond to information shocks faster than other coins. As a result, lagged returns of BTC serve as predictors of other coins' returns in the market. Again, the positive coefficient estimates suggest a faster response of their prices to common information shocks compared with other cryptocurrencies. Many coins show estimated coefficients with both positive and negative signs, a finding that is consistent with the complicated interdependencies across assets documented in Rapach, Strauss, Tu, and Zhou (2019). Moreover, the cryptocurrencies that serve as powerful predictors, such as BTC, will capture the positive lead-lag effect and the associated positive cross-cryptocurrencies predictability while the less powerful predictors, like those smaller coins, will capture the complicated interdependencies, which may lead to negative cross-cryptocurrencies predictability.

3.3. Principal component analysis

To further investigate the information spillover effect, we employ the principal component analysis (PCA), where the number of latent principal components that are common across cryptocurrencies is selected based on the Bai and Ng (2002) criteria. Specifically, we include the

lagged estimated factors as regressors as follows:

$$r_{i,t+1} = \alpha_i + \sum_{k=1}^{K} \beta_{i,k} \hat{f}_{k,t} + \epsilon_{i,t+1} \quad (6)$$

where the estimated factors $\hat{f}_{k,t}$ is the principal component estimate of a K-vector of latent factors that are common across cryptocurrencies. In our case the number K is set to be 3.

Figure 1 shows the estimated loadings for each cryptocurrency on the three principal components using a rolling window, with estimated factors standardized. All cryptocurrencies in our sample have positive and relatively uniform loadings on the first principal component, suggesting that common links among cryptocurrencies serve as the most important channel that contributes to the cross-cryptocurrency return predictability. Exceptions are again the three stablecoins (PAX, TUSD and USDC), which exhibit much smaller loadings due to their unique designs. In contrast to the uniform loadings on the first principal component, the loadings on the second and the third principal components fluctuate a lot across sample coins. The magnitude of loadings on the second principal component is close to zero for most coins, except for stablecoins. The fluctuating loadings on these two principal components are consistent with the finding in Table A.IV that are indicative of the complicated interdependencies between coins.

Overall, the uniform loadings on the first principal component indicate the presence of common shock serves as the most important channel of cross-cryptocurrency return predictability, while the varying loadings on the other two principal components capture some complex coin interdependencies in the cryptocurrency market. According to the return variance's distribution among the three main principal components, the first principal component represents 36.80% of the total variance, whereas the next two principal components together represent just 8.26%, indicating that the common links among cryptocurrencies have the highest explanation power for cross-cryptocurrency return predictability.

< To insert Figure 1 here. >

While Figure 1 provides an overall view about the weight for the entire sample period, there may exist potential time variations that arise due to the evolving dynamics of the cryptocurrency market. To address this concern, we examine the time-series weight of the first component for both the top three coins with the highest weight (BTC, ETH, and LTC) and the bottom three coins with the lowest weight (PAX, TUSD, USDC — which are the stable coins in our sample). The weights over time are plotted in Figure 2. We find that the weights for the top three coins experience a notable decline in early 2020, potentially influenced by the global economic disruptions stemming from the COVID-19 pandemic. Another significant drop in weights can be observed in late 2019, which we attribute to shifting policies in the US and China regarding cryptocurrency and blockchain technology, as in Table A.VII. That said, it is noteworthy that the weight variations, though present, are not drastic. Specifically, the weights for the top three coins consistently range between [0.2, 0.4], and these coins invariably maintain the highest weights throughout our sample period. Similarly, the weights for the bottom three coins stay within [0.0, 0.2] and consistently occupy the lowest weight bracket. The relative stability of these weights over time underscores the robustness of our PCA results.

< To insert Figure 2 here. >

3.4. Trading strategies

Given the strong cross-cryptocurrency return predictability, we next set about implementing profitable trading strategies and measuring their out-of-sample performance. The goal is to better align the aforementioned cross-cryptocurrency predictability with the utility of investor and to shed light on its economic value. We construct three long-short portfolios using out-of-sample minute-level cryptocurrency return forecasts predicted by the adaptive LASSO,

PCA, and the univariate (BTC) regression, respectively.

For the portfolios constructed based on the adaptive LASSO technique, we use half-day (720 minutes) data to forecast returns for each sample coin in the next minute, on a rolling basis. For each sample coin, i, we use data in the first 720 minutes on 25 March 2019 (referred to as "the first day" subsequently) to estimate $\beta_{i,j}$ via the adaptive LASSO technique in Equation (3). Next, we use the estimated parameters and data in the 720th minute on the first day to calculate the out-of-sample forecasts. In this way we generate a set of 30 excess return forecasts for the 721st minute on the first day. We then repeat this calculation for each minute level on a rolling basis. We sort the sample coins in ascending order based on the excess return forecasts and construct equal-weighted quintile portfolios on a minute frequency, and then form a long-short portfolio that goes long the top quintile portfolio and goes short the bottom quintile portfolio. To further test the robustness of our findings, we also construct portfolios using out-of-sample coin excess return forecasts based on the PCA approach. Again, we follow the aforementioned out-of-sample rolling-window estimation when applying the PCA prediction model shown in Equation (6).

As shown in Table II, BTC serves a market-driving role, where a one standard deviation increase in its lagged return will augment other coins' average return by 1.92 bps on a minute basis. Therefore, we construct another long-short portfolio using out-of-sample cryptocurrency return forecasts based on:

$$\mathrm{r}_{\mathrm{i,t+1}} = \hat{\alpha}_{i,t+1} + \hat{\beta}_{i,BTC}\, r_{BTC,t} + \epsilon_{i,t+1} \tag{7}$$

where $\mathrm{r}_{\mathrm{i,t+1}}$ is the return of cryptocurrency i at minute t+1, $r_{BTC,t}$ is the return of BTC at time t.

The portfolio construction method here is the same as that in the adaptive LASSO prediction, except that we now employ the prediction model in Equation (7). Using forecasted returns for each coin during the sample period, we sort coins into five equal-weighted quintiles

and construct a zero-investment portfolio that goes long (short) the top (bottom) quintile portfolio.

Table IV reports the out-of-sample returns of all constructed portfolios and the corresponding statistical significance. For all three prediction models, returns increase monotonically from the lowest quintile to the highest quintile. The minute-frequency return based on the adaptive LASSO forecasts is -1.61 bps in the lowest quintile portfolio, increasing markedly to 1.74 bps in the highest quintile. The return spread for the corresponding long-short portfolio is 3.34 bps at the 1% significance level. As for the PCA prediction method, the portfolio return climbs from -0.87 bps to 0.97 bps from the lowest quintile to the highest quintile. Apart from using the lagged returns of multiple coins as in-sample data, we report out-of-sample portfolio returns by using only BTC returns as regressors, according to Equation (7). Consistent with findings in Table II, BTC returns still serve as a strong predictor for other coins' returns. Specifically, the return spread for the long-short portfolio is 1.54 bps per minute at the 1% significance level. We also report the portfolio returns predicted via the adaptive LASSO and PCA by excluding the returns of the focal coins, and the long-short portfolio returns from using these two prediction techniques are 0.80 bps and 1.59 bps per minute, respectively. Both coefficient estimates remain statistically significant. Overall, results from Table IV serve as solid out-of-sample evidence of cross-cryptocurrency return predictability. More importantly, multiple trading strategies based on this spillover effect yield superior out-of-sample performance, indicating potential investment opportunities to crypto traders.

< To insert Table IV here. >

To ensure the results are not simply due to chance, we further conduct permutation tests (randomization) as a robustness check, specifically in accordance with the long-short portfolio return via the adaptive LASSO approach in Table IV. Firstly, we randomly select 60%, 70%, 80%,

or 90% of the data (in terms of the sample periods) and repeat the process 500 times. We depict the frequency distribution of the corresponding long-short portfolio returns in Figure A.2 Panel A. In all scenarios, the portfolio returns cluster around 0.0333 percent (3.33bps), very close to the 3.34 bps we documented in Table IV. Next, we randomly select 20 or 25 coins in our sample 100 times and report the frequency distribution of the portfolio return in Panel B. Again, the returns are close to 3.34 bps. Findings of the permutation tests further validate our main results.

3.5. Economic Channels

3.5.1. Risk-based explanation

One alternative potential explanation of the cross-cryptocurrency return predictability could be risk-based. We therefore test whether exposures to cryptocurrency risk factors can account for the performance of our trading strategies. Specifically, we control for multiple factors in the regression to check whether the alpha of the above strategies remain significant:

$$\mathrm{r_t} = \alpha_t + \beta_{MKT}\, MKT_t + \beta_{SMB}\, SMB_t + \beta_{MOM5}\, MOM5_t + \beta_{MOM10}\, MOM10_t + \beta_{MOM30}\, MOM30_t + \epsilon_t \quad (8)$$

where r_t is the portfolio return spread based on the adaptive LASSO prediction, MKT_t is the market factor, SMB_t is the "small-minus-big" size factor, and $MOM5_t$ is the "up-minus-down" momentum factor constructed based on the past 5-minute cumulative returns ($MOM10_t$ based on past 10-minute cumulative returns and $MOM30_t$ based on past 30-minute cumulative returns). Specifically, the risk-adjusted portfolio returns are minute-level, with available cryptocurrency data to construct risk factors. As for the market factor, we use sample cryptocurrencies' market capitalization in the prior day from *CoinMarketCap* as weights to calculate the value-weighted cryptocurrency market return. We sort cryptocurrency returns based on prior day's market capitalization to construct the "small-minus-big" size premium. We further use the minute-frequency cryptocurrency returns from *Binance* to calculate cumulative returns, and construct "up-

minus-down" momentum premiums based on the past 5-, 10-, or 30-minute cumulative returns, corresponding to factors $MOM5_t$, $MOM10_t$, $MOM30_t$, respectively.

Table V reports risk-adjusted alphas using single- or multiple-factor regressions indicated in Equation (8). The first five columns present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). The last column presents alphas controlling for all five factors in the regression. The long-short portfolio returns based on the adaptive LASSO forecasts remain significant and large, even with controlling for all risk factors. The long-short risk-adjusted alpha in the last column is 3.23 bps and is only slightly reduced from 3.34 bps shown in Table IV. The results reveal that the aforementioned return of the long-short portfolio cannot be explained by exposures to a variety of cryptocurrency risk factors.

< To insert Table V here. >

3.5.2. Investor attention channel

To further verify the information spillover hypothesis that is based on investors' limited attention, we use several proxies for investor attention and conduct sub-sample analysis. Cryptocurrency size serves as the first proxy. Research works such as Bamber (1987) and Zhang (2006) use firm size to represent information environment or the level of investor attention the firm can attract. They show that larger firms exhibit faster information diffusion compared to smaller firms. In our setting, we use the prior day's coin market capitalization as the proxy for the coin size. For each day, we divide the sample into terciles based on coin size and define coins in the top (bottom) tercile as large (small) coins. We then re-examine the risk-adjusted alphas of the portfolios constructed according to the adaptive LASSO prediction in each sub-sample.

The risk-adjusted alphas calculated from the factor regression in Equation (8) in large and small size groups are presented in the first two panels of Table VI. Panel A reports alphas for the

large-cap coins and Panel B shows results for the small-cap coins. The long-short portfolio alphas in both sub-samples with different regression specifications all remain significant. The risk-adjusted alpha with all risk factors controlled for the long-short portfolio is 3.75 bps for small coins (with low attention) and 1.65 bps for large coins (with high attention). The differences between risk-adjusted alphas from the two panels remain statistically significant, shown in Panel C. Hence, the higher return for coins with smaller size than for coins with larger size is consistent with slow information diffusion among cryptocurrencies due to investors' limited attention.

We also use abnormal Google search volume to proxy for investor attention.[21] By dividing the sample into terciles based on abnormal Google search volume, we report the risk-adjusted alphas calculated from the factor regression in Equation (8) in these sub-samples in Panel D – F of Table VI. Again, we define coins in the top tercile as coins with high abnormal Google search volume and those in the bottom tercile as coins with low abnormal Google search volume. Panel D and E report alphas for the coins with high abnormal Google search volume and low abnormal Google search volume, respectively. Although the long-short portfolio alphas in both sub-samples with different regression specifications remain significant, the risk-adjusted alpha with various risk factors controlled for the long-short portfolio is 3.29 bps for coins with low Google search volume (with low attention), significantly larger than the 2.80 bps for coins with high Google search volume (with high attention).[22]

Overall, the results in Table VI are consistent with slow information diffusion for cryptocurrencies resulting from investors' limited attention. Specifically, information diffuses at

[21] To obtain the abnormal Google search volume, we divide each coin's daily search volume by its average search volume in the past week and then subtract one, in accordance with the finance literature (e.g., Da, Engelberg, and Gao, 2011; Liu and Tsyvinski, 2021).

[22] The difference between risk-adjusted alphas for the sub-samples divided by abnormal Google search volume is statistically significant, as shown in Panel F.

a higher speed among coins that attract more investors' attention and thus common information is more promptly reflected in their prices. As a result, the return predictability based on peer coins' past returns becomes weaker for coins with larger investor attention, reinforcing the information spillover hypothesis.

< To insert Table VI here. >

For robustness, we use another measure for investor attention, which is distraction, following Eisensee and Strömberg (2007). Specifically, we calculate a daily distraction index from the average of the daily news pressure index based on the number of minutes that U.S. news broadcasts. We divide the whole sample into terciles based on the distraction index and treat the top tercile as high distraction periods and the rest as low distraction periods. Accordingly, high (low) distraction indicates low (high) attention. The results in Table VII are consistent with our main results, where the cross-crypto return predictability becomes weaker for coins with lower distraction, i.e., higher investor attention.

< To insert Table VII here. >

The analysis above provides empirical evidence that return predictability is stronger in a poorer information environment. Nonetheless, the contemporaneous correlation are expected to be weaker, according to Dou and Wu (2023). Therefore, we further test the investor attention channel by examining the contemporaneous correlation, using Google search volume and size to proxy for investor attention. Motivated by the methodology in Dou and Wu (2023), we divide the sample into terciles based on the attention proxies, and use the interaction between each tercile portfolio return forecast and return forecasts generated via adaptive LASSO as predictors, in both contemporaneous and predictive Fama-MacBeth regressions. Our findings, presented in Table VIII, indicate positive beta estimates for predictive regressions, whereas the estimates for

contemporaneous regressions are negative. For both Google search volume and size proxies, in the lowest tercile or in a poorer information environment, the contemporaneous correlation is weaker (more negative) while the return predictability is stronger, consistent with our expectation.

< To insert Table VIII here. >

The negative estimates for the contemporaneous correlation might stem from the competition of fund flows in the crypto markets. When limited funds flow into the crypto sphere, a surge in funds towards other coins, resulting in heightened returns, could lead to a relative decrease in funds and returns for the focal coin. This interplay manifests as a negative correlation between returns of the focal coin and its peers. For a poorer information environment, the focal coin faces more fluctuations and competition from fund flows, thereby intensifying the negative correlation observed.

Moreover, we examine the return spread of the long-short portfolio predicted via the adaptive LASSO approach, constructed in section 3.4, over the subsequent 10-minute horizon. The cumulative excess returns up to the next 10 minutes are presented in Figure 3. The return spread at time t+1 (the first minute) based on the adaptive LASSO prediction is sizable and significant[23] at 3.34 bps, as also shown in Table IV. Returns in the subsequent five minutes remain positive and statistically significant though the magnitude decreases over time. Returns after the sixth minute become insignificant, indicating that there is a slow information diffusion as cryptocurrency prices incorporate the information shocks gradually.

< To insert Figure 3 here. >

3.5.3. Liquidity channel

Besides the risk-based explanation, it is also possible that heterogeneous market liquidity of

[23] The t-statistics are not reported here for brevity.

coins affects the cross-crypto return predictability. For instance, coins with insufficient liquidity may respond to common shocks in a slower speed, and their returns can be predicted using the returns of coins with higher liquidity. To rule out this explanation, we use two proxies for coin liquidity and further conduct the Fama-Macbeth regression of the minute-level crypto returns. Specifically, we use Amihud's (2002) illiquidity and abnormal trading volume as two measures of coin liquidity and create two dummy variables accordingly. The dummy variables equal to 1 if the coin is in top tercile portfolio sorted by Amihud illiquidity or abnormal trading volume, and 0 otherwise. Another key variable is the out-of-sample forecast return predicated via adaptive LASSO. We include the interaction terms between the forecast return and each liquidity dummy, and report the regression results in Table IX. The liquidity measures as well as their interactions with the LASSO-forecasted returns indeed yield significant return predictability, indicating the plausible liquidity channel explanation. Still, we find that the adaptive LASSO methods maintain significant predictability after controlling for both liquidity measures and the interactions. In column (4) where both liquidity proxies are included, the return forecast generated by the adaptive LASSO method yields an estimated coefficient of 1.44, much higher than those of the liquidity measures. Overall, the above results show that the liquidity channel is consistent with information diffusion story but our results cannot be subsumed by the market liquidity explanation.

< To insert Table IX here. >

For robustness, we conduct sub-sample portfolio analyses by dividing the sample based on the Amihud illiquidity of coins and examine the risk-adjusted alphas. Table X reports the result. We observe that while the differences in risk-adjusted alphas between high liquidity and low liquidity samples are negative and significant, the alphas remain significantly positive for both sub-samples. These findings align with the regression results in Table IX, indicating that the

liquidity channel explanation does not subsume our main results.

< To insert Table X here. >

3.5.4. Market-wide events and information spillover hypothesis

Regarding the information-spillover hypothesis, we investigate nine market-wide events and examine the corresponding long-short portfolio alphas on these event days. These events are selected because of their broad impact on the cryptocurrency market, including President Xi's advocacy of blockchain in China, PayPal's adoption of cryptocurrencies as a payment tool, the IPO of the largest cryptocurrency platform Coinbase, the $3 billion PlusToken Ponzi scheme, and the "Black Thursday"[24] when the cryptocurrency market suddenly collapsed because of the spread of COVID-19.[25] Having had a broad impact on the overall cryptocurrency market, these events should affect various coins as common shocks. The long-short portfolio alphas on event days and normal days are reported in Table A.VIII, where we classify the days other than the nine event days in our sample as normal days. The risk-adjusted alphas are expected to be higher on the event days than on the normal days, as the market-wide events create abnormally larger common shocks to various coins and a consequential higher level of cross-cryptocurrency return predictability on the event days than on the normal days. Indeed, the risk-adjusted alpha with all risk factors controlled for the long-short portfolios on the event days (Panel A) is 4.15 bps, larger than the 3.21 bps on normal days (Panel B).[26] The differences between alphas on event days and normal days

[24] Black Thursday, while not directly linked to crypto-specific information, may amplify information asymmetry between larger and smaller coins. Excluding this event, the results for the remaining eight events show consistent but weaker positive alphas.

[25] A full list of the market events mentioned in this paper is provided in Table A.VII in the Appendix.

[26] There are two potential explanations of why the alphas on normal days remain significantly positive. One is that, on normal days, there could exist sporadic event-based common shocks other than the nine major events. Another is that there may be non-event-based common information shocks, such as an increasing number of digital wallet users or blockchain-related network activities in the market, given the relatively early stage of the cryptocurrency market. On normal days, when these common shocks occur, their impact may not be promptly reflected in prices.

are shown in Panel C and are statistically significant.

3.5.5. Varying market conditions

We have ruled out the alternative explanations of the cross-crypto predictability such as risk channel and liquidity channel. Still, varying crypto market conditions may affect the predictability. Kyle and Xiong (2001) point out the wealth effect where during the crypto market boom periods, the additional resources will be allocated to less competitive coins other than BTC and the coin prices will increase as a result. With regard to this possibility, we use CRIX, a market-level cryptocurrency index developed by Trimborn and Härdle (2018), to proxy for the crypto market conditions. We divide the whole sample periods into terciles based on the daily CRIX index and identify days in the top tercile as market boom days and the rest as normal days. The risk-adjusted alphas for market boom and normal days are reported in Table XI. In either sub-sample, the alphas remain significant. The differences of alphas between market boom and normal days are mostly insignificant. The findings here indicate that the wealth effect does not drive our main results.

< To insert Table XI here. >

Literature also documents that market uncertainty affects the connectedness and spillover effect among cryptocurrencies (e.g., Antonakakis, Chatziantoniou, and Gabauer, 2019; Al-Shboul, Assaf, and Mokni, 2022). To rule out the possibility that the cross-crypto return predictability is driven by varying market uncertainty among coins, we conduct sub-sample analysis of the risk-adjusted alphas in different market uncertainty environments. We use the crypto policy uncertainty (CCPO) measure proposed by Lucey, Vigne, Yarovaya, and Wang (2021)[27] and divide the whole sample into terciles based on CCPO. We treat the top tercile as high uncertainty days and the rest

[27] Lucey et al. (2021) develop two cryptocurrency uncertainty indices: CCPO and crypto price uncertainty (CCPR). The two indices have an extremely high correlation of 0.99 in our sample period. We use the former one to proxy for uncertainty.

as low uncertainty days and report the risk-adjusted portfolio alphas in Table XII. We find that the alphas in either sub-sample remain significant and the differences between two sub-samples is insignificant. The results indicate that the market uncertainty does not drive cross-crypto return predictability.

< To insert Table XII here. >

3.5.6. Robustness check

So far, we have examined the alternative channels separately. To comprehensively control for all three channels simultaneously, we apply a methodology inspired by Stambaugh and Yuan (2017). This involves using cryptocurrency return forecasts, generated via the adaptive LASSO method, to construct minute-level long-short portfolio returns. We then perform time-series regressions of these portfolio returns against proxies for liquidity, market wealth, and market uncertainty. Specifically, we include dummy variables for market wealth and market uncertainty – based on the CRIX and CCPO indices, respectively – and the return of the long-short portfolio constructed from Amihud illiquidity (as detailed in Table X). Table XIII reports the results. The positive and significant intercepts across all regression models suggest that none of the three channels fully account for the predictability of the crypto return forecasts we constructed.

< To insert Table XIII here. >

We further conduct the Granger causality test to corroborate the causal relationship underlying the spillover channel. Specifically, we use daily Google search volume to proxy for investor attention. Another key variable is the cumulative daily returns of minute-level portfolio returns constructed via the adaptive LASSO method. The results are reported in Table XIV. In Panel B, the null hypothesis for which Google search volume does not Granger-cause portfolio return of adaptive LASSO is rejected while the opposite hypothesis remains true. In other words,

investor limited attention indeed drives the cross-crypto return predictability.

< To insert Table XIV here. >

In addition, we check if the cross-crypto return predictability is driven by any pre-existing correlations or spurious relationships. To account for any inherent correlations, we adopt the average correlation method proposed by Pollet and Wilson (2010). Specifically, we calculate the cross-sectional average of the pairwise correlation of returns for all pairs among our 30 sample coins in a half-day rolling window. We assign the coins into terciles where we treat the top tercile as periods of high correlation and the remaining as periods of low correlation. The results, presented in Table XV, show that irrespective of being in a high or low correlation period, the portfolios consistently produce significantly positive alphas exceeding 3 bps. The differences in alphas between the two periods are both statistically insignificant and minimal, underscoring that pre-existing correlations do not drive the observed predictability.

< To insert Table XV here. >

We also address the potential concern of spurious relationships between the coin returns, for which we conduct the Augmented Dickey-Fuller (ADF) test[28]. Our results, shown in Table A.IX, consistently yield the ADF-stats with p-values of 0.01 for each coin, confirming the stationarity of our data and effectively ruling out spurious relationships.

3.6. Transaction costs

3.6.1. Trading strategies

The profits documented for the strategies outlined above may subject to relatively high transaction costs in the cryptocurrency spot market, which is 10 bps according to *Binance*. Also,

[28] The ADF test assesses the stationarity of the time-series data, ensuring our findings are not influenced by spurious predictive regressions.

we cannot short cryptos in the spot markets. Therefore, even if the aforementioned long-short investment strategies still yield sizable profits after considering the relatively high transaction costs, it may not be feasible in practice. Therefore, we further examine the portfolio performance when investors are trading in the futures contracts market, where the transaction cost for regular users is 4 bps for market takers[29].

3.6.2. Portfolio performance and transaction costs

We examine the out-of-sample performance using the adaptive LASSO strategy in the spot and futures markets. According to *Binance*, trading fees of futures contracts for makers are much lower than those for takers. We present the results in Table XVI using half-day and one-day as rolling windows and only report for regular takers to be conservative.[30]

Overall, after taking trading costs into account, the long-short portfolios yield statistically and economically significant returns for both spot and futures markets. According to results in Table XVI, with the relatively high trading costs of 12 bps in the spot market[31], the long-short portfolio still generates a statistically significant return of 0.74 bps after accounting for the trading costs with a half-day rolling window. However, it is noteworthy that such trading strategy is not feasible in practice for the spot market due to short sale constraint.

< To insert Table XVI here. >

As for the futures trading, the profits of the long-short portfolios for takers using quintile sorting are positive after considering transaction costs. For example, the long-short portfolio with transaction costs of 4 bps still generates a sizable return of 0.34 bps using a half-day rolling

[29] In the futures market, taker (maker) is an order that trades at a market (limited) price.
[30] The profits for makers would be higher under the same conditions.
[31] We follow Jia, Wu, Yan, and Liu (2023) to use 12 bps as the total transaction cost for the spot market, including bid-ask spread and trading fees.

window with 10-minute rebalancing[32]. The return after transaction costs remains significant at 1% level and the results are similar using a one-day rolling window. Also, the results remain robust in both spot and futures markets even after accounting for transaction costs if we use 1-minute return forecasts and rebalance every five minutes (reported in Appendix Table A.X). Hence, the statistically and economically significant long-short portfolio returns validate our prior findings of utilizing cross-cryptocurrency return predictability to achieve sizable profits.[33]

4. Conclusion

In this paper, we find strong cross-cryptocurrency return predictability, where the lagged returns of other cryptocurrencies significantly predict returns of the focal ones. In particular, playing a leading role in the cryptocurrency market, Bitcoin tends to respond to information more promptly and thus its lagged return is a strong predictor for other cryptocurrencies. We also employ the adaptive version of LASSO and the PCA approach to identify and include the lagged returns of other important coins beyond BTC as additional predictors. The results are all significant.

We attribute such findings to common shocks and the information spillover effect in the cryptocurrency market, given that most cryptocurrencies share similar underlying technical details. Due to the limited attention or constrained information-processing capabilities of investors, the information about common shocks is not homogeneously incorporated into individual coins, so the lagged returns of other coins may contain valuable information that is

[32] Because minute-level rebalancing is too costly and infeasible in practice, we use a ten-minute rebalancing interval.
[33] In practice, such high frequency trading may not be immediately executed so that the high frequency trading profits could be deteriorated in practice.

incorporated into the prices of the focal coins with a delay, thereby engendering cross-cryptocurrency return predictability. We run various tests to show that the slow information diffusion mechanism, rather than any risk-based ones or the liquidity channel, is more consistent with the cross-cryptocurrency return predictability. Further robustness tests show that our results are not driven by the wealth effect, market uncertainty, pre-existing correlations or spurious relationships.

In addition, we construct a long-short portfolio based on the past returns of cryptocurrencies in the spot and futures market and our portfolio can generate statistically and economically significant returns out-of-sample after accounting for transaction costs. This result indicates that using cross-cryptocurrency return predictability can achieve sizable economic gains.

References

Al-Shboul, Mohammad, Ata Assaf, and Khaled Mokni. 2022. "When Bitcoin Lost Its Position: Cryptocurrency Uncertainty and the Dynamic Spillover Among Cryptocurrencies Before and During the COVID-19 Pandemic." *International Review of Financial Analysis* 83: 102309.

Ali, Usman, and David Hirshleifer. 2020. "Shared Analyst Coverage: Unifying Momentum Spillover Effects." *Journal of Financial Economics* 136 (3): 649–675.

Amihud, Yakov. 2002. "Illiquidity and Stock Returns: Cross-Section and Time-Series Effects." *Journal of Financial Markets* 5 (1): 31–56.

Antonakakis, Nikolaos, Ioannis Chatziantoniou, and David Gabauer. 2019. "Cryptocurrency Market Contagion: Market Uncertainty, Market Complexity, and Dynamic Portfolios." *Journal of International Financial Markets, Institutions and Money* 61: 37–51.

Badrinath, S.G., Jayant R. Kale, and Thomas H. Noe. 1995. "Of Shepherds, Sheep, and the Cross-autocorrelations in Equity Returns." *Review of Financial Studies* 8 (2): 401–430.

Bai, Jushan and Serena Ng. 2002. "Determining the Number of Factors in Approximate Factor Models." *Econometrica* 70 (1): 191–221.

Bamber, Linda Smith. 1987. "Unexpected Earnings, Firm Size, and Trading Volume around Quarterly Earnings Announcements." *Accounting Review* 62 (3): 510–532.

Brauneis, Alexander, and Roland Mestel. 2018. "Price Discovery of Cryptocurrencies: Bitcoin and Beyond." *Economics Letters* 165: 58–61.

Brennan, Michael J., Narasimhan Jegadeesh, and Bhaskaran Swaminathan. 1993. "Investment Analysis and the Adjustment of Stock Prices to Common Information." *Review of Financial Studies* 6 (4): 799–824.

Campbell, Harvey. 2017. "Presidential Address: The Scientific Outlook in Financial Economics."

Journal of Finance 72 (4): 1399–1440.

Campbell, John Y., Sanford J. Grossman, and Jiang Wang. 1993. "Trading Volume and Serial Correlation in Stock Returns." *Quarterly Journal of Economics* 108 (4): 905–939.

Chaim, Pedro and Márcio P. Laurini. 2019. "Nonlinear Dependence in Cryptocurrency Markets." *North American Journal of Economics and Finance* 48: 32–47.

Chen, Wen, Bo Hu, and Liyan Yang. 2022. "Whence LASSO? A Rational Interpretation." Working Paper.

Chinco, Alex, Adam D. Clark-Joseph and Mao Ye. 2019. "Sparse Signals in the Cross-Section of Returns." *Journal of Finance* 74 (1): 449–492.

Chohan, Usman W. 2021. "Are Stable Coins Stable?" Working Paper.

Chordia, Tarun, and Bhaskaran Swaminathan. 2000. "Trading Volume and Cross-Autocorrelations in Stock Returns." *Journal of Finance* 55 (2): 913–935.

Ciaian, Pavel, Miroslava Rajcaniova, and d' Artis Kancs. 2018. "Virtual Relationships: Short-and Long-Run Evidence from Bitcoin and Altcoin Markets." *Journal of International Financial Markets, Institutions and Money* 52: 173–195.

Cohen, Lauren, and Andrea Frazzini. 2008. "Economic Links and Predictable Returns." *Journal of Finance* 63 (4): 1977–2011.

Cong, Lin William and Zhiguo He. 2019. "Blockchain Disruption and Smart Contracts." *Review of Financial Studies* 32 (5): 1754–1797.

Cong, Lin William, Zhiguo He and Jiasun Li. 2021. "Decentralized Mining in Centralized Pools." *Review of Financial Studies* 34 (3): 1191–1235.

Corbet, Shaen, Andrew Meegan, Charles Larkin, Brian Lucey and Larisa Yarovaya. 2018. "Exploring the Dynamic Relationships between Cryptocurrencies and Other Financial Assets."

Economics Letters 165: 28–34.

Da, Zhi, Joseph Engelberg, and Pengjie Gao. 2011. "In Search of Attention." *Journal of Finance* 66 (5):1461–1499.

Da, Rui, Stefan Nagel, and Dacheng Xiu. 2022. "The Statistical Limits of Arbitrage." Working Paper.

Detzel, Andrew, Hong Liu, Jack Strauss, Guofu Zhou, and Yingzi Zhu. 2020. "Learning and Predictability via Technical Analysis: Evidence from Bitcoin and Stocks with Hard-to-value Fundamentals." *Financial Management* 50 (1): 107–137.

Dou, Winston Wei and Wei Wu. 2023. "Competition Network and Predictable Industry Returns." Working Paper.

Duffie, Darrell. 2010. "Presidential Address: Asset Price Dynamics with Slow-Moving Capital." *Journal of Finance* 65 (4): 1237–1267.

Eisensee, Thomas and David Strömberg. 2007. "News Droughts, News Floods, and US Disaster Relief." *Quarterly Journal of Economics* 122: 693–728.

Freyberger, Joachim, Andreas Neuhierl and Michael Weber. 2020. "Dissecting Characteristics Nonparametrically." *Review of Financial Studies* 33 (5): 2326–2377.

Fry, John and Eng-Tuck Cheah. 2016. "Negative Bubbles and Shocks in Cryptocurrency Markets." *International Review of Financial Analysis* 47: 343–352.

Gu, Shihao, Bryan Kelly and Dacheng Xiu. 2020. "Empirical Asset Pricing via Machine Learning." *Review of Financial Studies* 33 (5): 2223–2273.

Hendershott, Terrence and Albert J. Menkveld. 2014. "Price Pressures." *Journal of Financial Economics* 114 (3): 405–423.

Heston, Steven L., Robert A. Korajczyk, and Ronnie Sadka. 2010. "Intraday Patterns in the Cross-

section of Stock Returns." *Journal of Finance* 65 (4): 1369–1407.

Hoang, Lai T., and Dirk G. Baur. 2021. "How Stable Are Stablecoins?" *European Journal of Finance*, forthcoming.

Hong, Harrison, Terence Lim, and Jeremy C. Stein. 2000. "Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies." *Journal of Finance* 55 (1): 265–295.

Hong, Harrison, and Jeremy C. Stein. 1999. "A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets." *Journal of Finance* 54 (6): 2143–2184.

Hong, Harrison, Walter Torous, and Rossen Valkanov. 2007. "Do Industries Lead Stock Markets?" *Journal of Financial Economics* 83 (2): 367–396.

Hou, David, Jennifer Li, Li Liao, and Hong Zhang. 2019. "Asset Pricing on Blockchain: Slow Moving Capital, Momentum, and Bubbles of Cryptocurrencies." Working Paper.

Hou, Kewei. 2007. "Industry Information Diffusion and the Lead-lag Effect in Stock Returns." *Review of Financial Studies* 20 (4): 1113–1138.

Huang, Xiaoran, Juan Lin, and Peng Wang. 2022. "Are Institutional Investors Marching into the Crypto Market?" *Economics Letters* 220.

Jaffe, Adam B., Manuel Trajtenberg, and Rebecca Henderson. 1993. "Geographic Localization of Knowledge Spillovers as Evidenced by Patent Citations." *Quarterly Journal of Economics* 108 (3): 577–598.

Ji, Qiang, Elie Bouri, Chi Keung Marco Lau and David Roubaud. 2019. "Dynamic Connectedness and Integration in Cryptocurrency Markets." *International Review of Financial Analysis* 6: 257–272.

Jia, Yuecheng, Yangru Wu, Shu Yan, and Yuzheng Liu. 2023. "A Seesaw Effect in the

Cryptocurrency Market: Understanding the Return Cross Predictability of Cryptocurrencies." *Journal of Empirical Finance* 74: 101428.

Katsiampa, Paraskevi, Shaen Corbet and Brian Lucey. 2019. "Volatility Spillover Effects in Leading Cryptocurrencies: A BEKK-MGARCH Analysis." Finance Research Letters 29: 68–74.

Kyle, Albert S., Wei Xiong. 2001. "Contagion as a Wealth Effect." *Journal of Finance* 56 (4): 1401–1440.

Kozak, Serhiy, Stefan Nagel and Shrihari Santosh. 2020. "Shrinking the Cross-Section." Journal of Financial Economics 135 (2): 271–292.

Kraus, Alan, and Hans R. Stoll. 1972. "Price Impacts of Block Trading on the New YorkStock Exchange." *Journal of Finance* 27 (3): 569–588.

Lee, Charles M.C., Stephen Teng Sun, Rongfei Wang, and Ran Zhang. 2019. "Technological Links and Predictable Returns." *Journal of Financial Economics* 132 (3): 76–96.

Lee, Jongsub, Tao Li, and Donghwa Shin. 2021. "The Wisdom of Crowds in FinTech: Evidence from Initial Coin Offerings." *Review of Corporate Finance Studies*, forthcoming.

Li, Jiasun and Guanxi Yi. 2019. "Toward a Factor Structure in Crypto Asset Returns." *Journal of Alternative Investments* 21 (4): 56–66.

Liu, Jiatao, Ian W. Marsh, Paolo Mazza, and Mikael Petitjean. 2021. "Factor Structure in Cryptocurrency Return and Volatility." Working Paper.

Liu, Yukun, Jinfei Sheng, and Wanyi Wang. 2021. "Technology and Cryptocurrency Valuation: Evidence from Machine Learning." Working Paper.

Liu, Yukun and Aleh Tsyvinski. 2021. "Risks and Returns of Cryptocurrency." *Review of Financial Studies* 34 (6): 2689–2727.

Liu, Yukun, Aleh Tsyvinski, and Xi Wu. 2021. "Common Risk Factors in Cryptocurrency."

Journal of Finance, forthcoming.

Lo, Andrew W. and A. Craig MacKinlay. 1990. "When Are Contrarian Profits Due to Stock Market Overreaction?" *Review of Financial Studies* 3 (2): 175–205.

Lucey, Brian M., Samuel A. Vigne, Larisa Yarovaya, and Yizhi Wang. 2022. "The Cryptocurrency Uncertainty Index." *Finance Research Letters* 45: 102147.

Makarov, Igor and Antoinette Schoar. 2020. "Trading and arbitrage in cryptocurrency markets." *Journal of Financial Economics* 135 (2): 293–319.

Menzly, Lior and Oguzhan Ozbas. 2010. "Market Segmentation and Cross-predictability of Returns." *Journal of Finance* 65 (4): 1555–1580.

Merton, Robert C. 1987. "A Simple Model of Capital Market Equilibrium with Incomplete Information." *Journal of Finance* 42 (3): 483–510.

Moratis, George. 2021. "Quantifying the Spillover Effect in the Cryptocurrency Market." *Finance Research Letters* 38: 101534.

Nakamoto, Satoshi. 2008. "Bitcoin: A Peer-to-peer Electronic Cash System." Working Paper.

Omane-Adjepong, Maurice and Imhotep Alagidede. 2019. "Multiresolution Analysis and Spillovers of Major Cryptocurrency Markets." *Research in International Business and Finance* 49: 191–206.

Palamalai, Srinivasan and Bipasha Maity. 2019. "Return And Volatility Spillover Effects In Leading Cryptocurrencies." *Global Economy Journal* 19 (3): 1950017.

Petukhina, Alla A., Raphael C. G. Reule and Wolfgang Karl Härdle. 2021. "Rise of the Machines? Intraday High-Frequency Trading Patterns of Cryptocurrencies." *European Journal of Finance* 27 (1–2): 8–30.

Pollet, M. Joshua, and Mungo Wilso. 2010. "Average Correlation and Stock Market Returns."

Journal of Financial Economics 96 (3): 364–380.

Rapach, David, Jack Strauss, Jun Tu, and Guofu Zhou. 2019. "Industry Return Predictability: A Machine Learning Approach." *Journal of Financial Data Science* 1 (3): 9–28.

Rapach, David, Jack Strauss, and Guofu Zhou. 2013. "International Stock Return Predictability: What Is the Role of the United States?" *Journal of Finance* 68 (4): 1633–1662.

Shahrur, Husayn, Ying L. Becker, and Didier Rosenfeld. 2010. "Return Predictability along the Supply Chain: The International Evidence." *Financial Analysts Journal* 66 (3): 60–77.

Shams, Amin. 2020. "The Structure of Cryptocurrency Returns." Working Paper.

Sifat, Imtiaz Mohammad, Azhar Mohamad and Mohammad Syazwan Bin Mohamed Shariff. 2019. "Lead-Lag Relationship between Bitcoin and Ethereum: Evidence from Hourly and Daily Data." *Research in International Business and Finance* 50: 306–321.

Sockin, Michael and Wei Xiong. 2021. "A Model of Cryptocurrencies." Working Paper.

Stambaugh, Robert F. and Yu Yuan. 2017. "Mispricing Factors." *Review of Financial Studies* 30 (4): 1270–1315.

Subramaniam, Sowmya and Madhumita Chakraborty. 2020. "Investor Attention and Cryptocurrency Returns: Evidence from Quantile Causality Approach." *Journal of Behavioral Finance* 21 (1): 103–115.

Symitsi, Efthymia and Konstantinos J. Chalvatzis. 2018. "Return, Volatility and Shock Spillovers of Bitcoin with Energy and Technology Companies." *Economics Letters* 170: 127–130.

Tibshirani, Robert. 1996. "Regression Shrinkage and Selection via the Lasso." *Journal of the Royal Statistical Society. Series B (Methodological)* 58 (1): 267–288.

Trimborn, Simon and Wolfgang Karl Härdle. 2018. "CRIX an Index for Cryptocurrencies." *Journal of Empirical Finance* 49: 107–122.

Wang, Gang-Jin, Chi Xie, Danyan Wen and Longfeng Zhao. 2019. “When Bitcoin Meets Economic Policy Uncertainty (EPU): Measuring Risk Spillover Effect from EPU to Bitcoin.” *Finance Research Letter* 31: 489–497.

Zhang, X. Frank. 2006. “Information Uncertainty and Stock Returns.” *Journal of Finance* 61 (1): 105–137.

Zou, Hui. 2006. “The Adaptive Lasso and Its Oracle Properties.” *Journal of the Royal Statistical Society. Series B (Methodological)* 101: 1418–1429.

Table I Summary statistics

Table I reports summary statistics for minute-level returns of the top 30 cryptocurrencies, which are selected based on both the large market capitalization and the length of the trading history. The sample period is from 25 March 2019 to 30 April 2021, and the data frequency is on a minute basis in the 24/7 cryptocurrency market. Returns are standardized with means equal to 0 and standard deviations equal to 1. Sharpe ratio is the average return of each coin in excess of the risk-free rate, divided by the coin's standard deviation. Data are obtained from *Binance*.

Symbol	Name	Mean (bps)	Standard deviation (%)	Minimum (%)	Maximum (%)	Sharpe ratio (%)
ADA	Ada	0.05	0.18	-7.64	8.35	0.25
BAT	Basic Attention Token	0.04	0.21	-12.34	10.20	0.21
BNB	Binance Coin	0.05	0.17	-9.81	10.37	0.29
BTC	Bitcoin	0.03	0.13	-7.24	7.50	0.27
BTT	BitTorrent	0.05	0.22	-13.23	15.71	0.22
CELR	Celer	0.05	0.31	-7.34	12.00	0.17
EOS	EOS coin	0.02	0.17	-6.28	13.90	0.12
ETC	Ethereum Classic	0.04	0.19	-5.73	8.62	0.23
ETH	Ether	0.04	0.15	-5.88	8.17	0.26
FET	Fetch.ai	0.04	0.29	-8.41	13.23	0.15
HOT	HoloTokens	0.08	0.29	-9.96	11.69	0.25
ICX	ICON	0.05	0.24	-10.96	17.35	0.21
IOST	IOStoken	0.05	0.23	-7.88	12.10	0.20
IOTA	IOTA tokens	0.04	0.20	-7.60	10.12	0.19
LINK	Link token	0.07	0.23	-14.51	10.16	0.28
LTC	Litecoin	0.03	0.17	-7.03	10.37	0.17
NEO	Neo token	0.04	0.18	-5.83	9.07	0.22
ONT	Ontology Coin	0.02	0.20	-8.36	6.48	0.12
PAX	Paxos Standard Token	0.00	0.02	-2.96	8.89	-0.01
QTUM	Qtum	0.04	0.19	-5.82	8.80	0.18
TRX	Tronix	0.03	0.17	-7.13	8.50	0.16
TUSD	TrueUSD	0.00	0.02	-1.95	2.34	0.00
USDC	USD Coin	0.00	0.02	-1.70	2.12	0.00
VET	VeChain Token	0.06	0.23	-9.38	22.94	0.25
WAVES	Waves	0.04	0.22	-6.23	17.35	0.16
XLM	Stellar Lumens	0.03	0.18	-9.47	6.53	0.18
XMR	Monero	0.03	0.16	-4.53	6.58	0.17
XRP	XRP coins	0.04	0.20	-15.30	12.45	0.18
ZEC	Zcash	0.03	0.19	-6.75	7.33	0.16
ZIL	Zilliqa	0.04	0.24	-8.08	15.23	0.18

Table II OLS predictive regression results: lagged BTC returns

Table II reports coefficient estimates for each of the 30 cryptocurrencies, with the lagged return of Bitcoin as the independent variable and its own lagged return as control variable. Return of each cryptocurrency shown in the table serves as the dependent variable in the OLS regression. Each value in the row (*btc_lag*) is the coefficient estimate of the lagged return of Bitcoin. Returns are on a minute basis and are standardized. Estimates are presented in bps. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	ADA	BAT	BNB	BTC	BTT	CELR
btc_lag	1.47***	2.58***	1.12***	–	2.24***	3.44***
	EOS	ETC	ETH	FET	HOT	ICX
btc_lag	0.73***	1.49***	0.76***	3.99***	3.50***	2.99***
	IOST	IOTA	LINK	LTC	NEO	ONT
btc_lag	2.93***	3.11***	2.08***	0.89***	1.95***	2.07***
	PAX	QTUM	TRX	TUSD	USDC	VET
btc_lag	-0.10***	2.39***	1.14***	-0.14***	-0.13***	2.58***
	WAVES	XLM	XMR	XRP	ZEC	ZIL
btc_lag	2.81***	1.33***	2.53***	0.67***	1.99***	3.29***

Table III Pooled OLS predictive regression results

Table III reports the pooled OLS coefficient estimates. For each cryptocurrency, the pooled OLS coefficient estimates of lagged returns of the other 29 cryptocurrencies are reported with the bias-corrected wild bootstrapped 95% confidence intervals shown in brackets. Estimates are presented in bps and those with bootstrapped p-values at the 5% level or better are in bold. The sample period is from 25 March 2019 to 30 April 2021.

ADA	BAT	BNB	BTC	BTT
0.91	-0.13	**2.27**	**4.82**	0.05
[0.42,1.40]	[-0.48,0.23]	[1.25,3.28]	[3.03,6.60]	[-0.26,0.36]
CELR	EOS	ETC	ETH	FET
-0.05	**1.45**	**0.65**	**2.27**	0.01
[-0.23,0.14]	[0.59,2.31]	[0.10,1.20]	[0.94,3.61]	[-0.21,0.23]
HOT	ICX	IOST	IOTA	LINK
-0.26	0.10	-0.08	-0.29	**0.63**
[-0.49,-0.02]	[-0.17,0.37]	[-0.50,0.34]	[-0.75,0.16]	[0.29,0.98]
LTC	NEO	ONT	PAX	QTUM
1.38	**1.04**	**0.80**	-1.56	**0.40**
[0.46,2.30]	[0.54,1.54]	[0.28,1.32]	[-6.25,3.12]	[0.06,0.74]
TRX	TUSD	USDC	VET	WAVES
1.85	2.93	-0.38	0.07	-0.02
[1.28,2.41]	[-2.92,8.78]	[-5.12,4.36]	[-0.20,0.34]	[-0.37,0.32]
XLM	XMR	XRP	ZEC	ZIL
1.04	-0.41	0.30	**1.05**	-0.02
[0.53,1.54]	[-0.90,0.08]	[-0.22,0.83]	[0.60,1.51]	[-0.30,0.27]

Table IV Out-of-sample portfolio performance

Table IV reports the out-of-sample performance of portfolios constructed using minute-level cryptocurrency return forecasts (in bps) predicted by the adaptive LASSO, PCA, and univariate (BTC) regression, respectively. For each minute, we sort the 30 sample coins in ascending order based on the excess return forecasts generated from each of the approaches using a half-day rolling window and group them into equal-weighted quintile portfolios. We subsequently construct a long-short portfolio that goes long the top quintile portfolio and goes short the bottom quintile portfolio, for each regression model. Minute-level portfolio returns are reported with the Newey-West adjusted t-statistics shown in the parentheses. The t-statistics of the minute-level portfolio returns are larger than three, thus meeting the requirement of the multi-hypothesis threshold posed in Harvey (2017). We also report the portfolio returns predicted via the adaptive LASSO and PCA by excluding the returns of the focal coins. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Adaptive LASSO	PCA	Bitcoin	Adaptive LASSO (ExFocal)	PCA (ExFocal)
Short	-1.61***	-0.87***	-0.73***	-0.33***	-0.75***
2	-0.28***	-0.34***	-0.29***	-0.11***	-0.34***
3	0.02	0.03**	0.03**	0.02	0.03**
4	0.32***	0.40***	0.36***	0.15***	0.41***
Long	1.74***	0.97***	0.81***	0.46***	0.84***
Long - Short	3.34***	1.85***	1.54***	0.80***	1.59***
T-stats	*(73.84)*	*(61.52)*	*(49.53)*	*(32.30)*	*(51.90)*

Table V Risk-adjusted portfolio alphas

Table V reports risk-adjusted alphas (in bps) based on single- or multiple-factor regressions using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) – (5) present alphas by controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (6) presents alphas by controlling for all five factors in the regression. Newey-West adjusted t-statistics are shown in the parentheses and are larger than three, thus meeting the requirement of the multi-hypothesis threshold posed in Harvey (2017). The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.64***	-1.62***	-1.74***	-1.79***	-1.82***	-1.62***
2	-0.31***	-0.29***	-0.58***	-0.58***	-0.55***	-0.46***
3	-0.01	0.01	-0.25***	-0.25***	-0.22***	-0.12***
4	0.29***	0.31***	0.06	0.07*	0.12***	0.20***
Long	1.71***	1.73***	1.43***	1.50***	1.62***	1.61***
Long - Short	3.35***	3.35***	3.17***	3.29***	3.44***	3.23***
T-stats	*(84.59)*	*(87.05)*	*(82.03)*	*(92.30)*	*(93.29)*	*(81.24)*

Table VI Sub-sample analysis

Table VI reports risk-adjusted alphas (in bps) for various sub-samples based on single- or multiple-factor regressions using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors. For each day, we divide the sample into terciles based on market capitalization (Panel A – B) and abnormal Google search volume (Panel D – E), respectively. The differences between risk-adjusted alphas for the sub-samples divided by either attention proxy are reported in Panel C and F. Newey-West adjusted t-statistics are shown in the parentheses. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A Large cap							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-0.72***	-0.75***	-0.73***	-1.05***	-1.05***	-1.02***	-0.88***
2	-0.14***	-0.18***	-0.16***	-0.47***	-0.47***	-0.43***	-0.31***
3	0.02	-0.01	0.01	-0.27***	-0.26***	-0.21***	-0.09***
4	0.20***	0.17***	0.18***	-0.09*	-0.07	-0.03	0.11***
Long	0.83***	0.80***	0.82***	0.53***	0.57***	0.65***	0.77***
Long - Short	1.55***	1.55***	1.55***	1.59***	1.62***	1.67***	1.65***
T-stats	*(63.23)*	*(56.92)*	*(55.80)*	*(57.40)*	*(57.44)*	*(54.27)*	*(48.23)*
Panel B Small cap							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.92***	-1.94***	-1.92***	-1.97***	-2.03***	-2.08***	-1.86***
2	-0.37***	-0.40***	-0.38***	-0.65***	-0.65***	-0.63***	-0.56***
3	0.02	0.00	0.01	-0.24***	-0.24***	-0.21***	-0.14***
4	0.40***	0.37***	0.39***	0.17***	0.18***	0.23***	0.29***
Long	2.05***	2.02***	2.04***	1.74***	1.82***	1.95***	1.89***
Long - Short	3.96***	3.96***	3.96***	3.70***	3.85***	4.04***	3.75***
T-stats	*(72.08)*	*(111.02)*	*(114.20)*	*(74.10)*	*(81.48)*	*(87.48)*	*(72.01)*

Table VI (Continued) Sub-sample analysis

Panel C Difference between large and small size

	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	1.20***	1.19***	1.19***	0.91***	0.98***	1.07***	0.98***
2	0.23***	0.22***	0.22***	0.17***	0.18***	0.21***	0.26***
3	0.00	-0.01	0.00	-0.03**	-0.02	-0.01	0.05***
4	-0.20***	-0.21***	-0.21***	-0.25***	-0.25***	-0.25***	-0.17***
Long	-1.21***	-1.23***	-1.23***	-1.21***	-1.25***	-1.30***	-1.12***
Long - Short	-2.41***	-2.42***	-2.42***	-2.12***	-2.23***	-2.37***	-2.10***
T-stats	*(-51.33)*	*(-50.43)*	*(-47.94)*	*(-42.79)*	*(-45.26)*	*(-45.13)*	*(-36.47)*

Panel D High abnormal Google search volume

	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.33***	-1.36***	-1.34***	-1.56***	-1.59***	-1.61***	-1.44***
2	-0.28***	-0.30***	-0.29***	-0.61***	-0.60***	-0.57***	-0.49***
3	0.00	-0.03***	-0.02	-0.30***	-0.30***	-0.26***	-0.16***
4	0.35***	0.32***	0.33***	0.06	0.08*	0.13***	0.22***
Long	1.45***	1.42***	1.44***	1.16***	1.21***	1.33***	1.36***
Long - Short	2.79***	2.79***	2.79***	2.72***	2.80***	2.94***	2.80***
T-stats	*(69.60)*	*(69.81)*	*(67.75)*	*(64.63)*	*(67.28)*	*(61.93)*	*(57.38)*

Panel E Low abnormal Google search volume

	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.68***	-1.71***	-1.69***	-1.77***	-1.82***	-1.86***	-1.65***
2	-0.30***	-0.33***	-0.31***	-0.57***	-0.58***	-0.56***	-0.46***
3	0.03*	0.00	0.02	-0.24***	-0.23***	-0.20***	-0.11***
4	0.35***	0.32***	0.34***	0.10**	0.11***	0.16***	0.24***
Long	1.79***	1.77***	1.79***	1.48***	1.55***	1.67***	1.64***
Long - Short	3.47***	3.47***	3.47***	3.24***	3.38***	3.53***	3.29***
T-stats	*(69.58)*	*(121.06)*	*(92.49)*	*(76.11)*	*(89.32)*	*(91.02)*	*(78.60)*

Panel F Difference between high and low abnormal Google search volume

	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	0.34***	0.34***	0.34***	0.21***	0.23***	0.25***	0.21***
2	0.03**	0.03*	0.03**	-0.03*	-0.02	0.00	-0.02
3	-0.03**	-0.03**	-0.03**	-0.06***	-0.07***	-0.05***	-0.05**
4	0.00	0.00	0.00	-0.03	-0.03	-0.02	-0.02
Long	-0.33***	-0.33***	-0.34***	-0.31***	-0.33***	-0.32***	-0.28***
Long - Short	-0.67***	-0.67***	-0.67***	-0.52***	-0.56***	-0.58***	-0.48***
T-stats	*(-16.85)*	*(-16.02)*	*(-14.87)*	*(-12.63)*	*(-13.68)*	*(-11.49)*	*(-9.68)*

Table VII Long-short portfolio performance for different distraction periods

Table VII reports risk-adjusted alphas (in bps) for different distraction periods using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors. We compute a daily distraction series following Eisensee and Strömberg (2007). We divide the whole sample into terciles based on distraction and treat the top tercile as high distraction days and the rest as low distraction days. Panel A (B) reports the risk-adjusted alphas for high (low) distraction periods and the differences between risk-adjusted alphas for the sub-samples are reported in Panel C. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: High distraction							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-2.64***	-2.42***	-2.50***	-2.81***	-2.80***	-2.82***	-2.49***
2	-0.66**	-0.45**	-0.51*	-0.87**	-0.86**	-0.93**	-0.59**
3	-0.31	-0.10	-0.16	-0.53	-0.52	-0.56	-0.25
4	0.04	0.26	0.19	-0.10	-0.15	-0.19	0.20
Long	1.95***	2.17***	2.11***	1.95***	1.92***	1.93***	2.33***
Long - Short	4.59***	4.59***	4.62***	4.76***	4.72***	4.75***	4.83***
T-stats	*(12.86)*	*(12.30)*	*(12.72)*	*(11.89)*	*(12.96)*	*(13.64)*	*(12.84)*
Panel B: Low distraction							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.61***	-1.63***	-1.62***	-1.73***	-1.79***	-1.82***	-1.62***
2	-0.28***	-0.31***	-0.29***	-0.58***	-0.58***	-0.55***	-0.46***
3	0.02	-0.01	0.01	-0.25***	-0.25***	-0.22***	-0.12***
4	0.32***	0.29***	0.31***	0.06	0.07*	0.13***	0.20***
Long	1.74***	1.71***	1.73***	1.43***	1.50***	1.62***	1.61***
Long - Short	3.34***	3.34***	3.34***	3.17***	3.29***	3.43***	3.22***
T-stats	*(85.57)*	*(84.46)*	*(86.41)*	*(81.71)*	*(91.49)*	*(92.71)*	*(81.03)*
Panel C: Difference between high and low distraction							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.04**	-0.71**	-0.90**	-1.03**	-0.99**	-0.99*	-0.70**
2	-0.38	-0.05	-0.23	-0.36	-0.31	-0.32	-0.02
3	-0.33	0.01	-0.17	-0.31	-0.26	-0.28	0.02
4	-0.28	0.05	-0.13	-0.27	-0.22	-0.24	0.06
Long	0.22	0.54*	0.35	0.23	0.27	0.24	0.54*
Long - Short	1.25***	1.26	1.25***	1.26***	1.26***	1.23***	1.25***
T-stats	*(2.86)*	*(2.85)*	*(2.86)*	*(2.92)*	*(2.86)*	*(2.76)*	*(2.87)*

Table VIII Contemporaneous correlation

Table VIII reports the Fama-MacBeth results for contemporaneous and predictive regressions. Minute-level cryptocurrency returns (in bps) are used as dependent variables. We divide the sample into terciles based on Google search volume (size) and use the interaction between each tercile portfolio return forecast and return forecasts generated based on adaptive LASSO as predictors in Panel A (Panel B). Tercile 3 (1) represents the top (bottom) tercile. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A Google search volume		
	Beta (contemporaneous)	Beta (predictive)
Adaptive LASSO×Tercile 1 (Low)	-221.55***	56.33***
	(-28.81)	(6.23)
Adaptive LASSO×Tercile 2	-191.26***	42.54***
	(-37.00)	(7.47)
Adaptive LASSO×Tercile 3 (High)	-168.60***	35.95***
	(-28.90)	(6.13)
Panel B Size		
	Beta (contemporaneous)	Beta (predictive)
Adaptive LASSO×Tercile 1 (Low)	-337.67***	64.67***
	(-152.68)	(113.15)
Adaptive LASSO×Tercile 2	-254.11***	59.66***
	(-88.12)	(90.82)
Adaptive LASSO×Tercile 3 (High)	-206.12***	54.94***
	(-61.64)	(63.51)

Table IX Liquidity and return predictability

Table IX reports the Fama-MacBeth regression results when taking liquidity into account. We use two liquidity proxies: Amihud's (2002) illiquidity (*Amihud*) and abnormal trading volume (*Abvol*). Minute-level cryptocurrency returns (in bps) are used as dependent variables. Predictors are the out-of-sample forecast returns based on adaptive LASSO, dummy variables that equal to 1 if the coin is in top tercile sorted by Amihud illiquidity and abnormal trading volume (and 0 otherwise). The two interactive terms are also included. The adaptive LASSO predictor is standardized cross-sectionally to have zero mean and unit standard deviation. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Return					
	(1)	(2)	(3)	(4)	(5)	(6)
Adaptive LASSO	1.45***	1.63***	1.44***	1.44***	1.11***	1.42***
	(224.55)	(222.43)	(223.43)	(221.44)	(107.01)	(183.69)
Amihud		0.04***		0.10***	0.04***	
		(5.79)		(12.42)	(5.61)	
Abvol			0.25***	0.26***		0.25***
			(32.83)	(32.42)		(30.49)
Amihud× Adaptive LASSO					0.48***	
					(37.28)	
Abvol×Adaptive LASSO						-0.15***
						(-9.73)
Intercept	0.04***	0.02**	-0.05***	-0.08**	0.02	-0.05***
	(2.80)	(1.75)	(-3.52)	(-6.28)	(1.58)	(-3.80)
R^2	0.08	0.12	0.12	0.16	0.17	0.17

Table X Long-short portfolio performance for different liquidity levels

Table X reports risk-adjusted alphas (in bps) for different liquidity levels using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors. We use (the inverse of) Amihud's (2002) illiquidity as the proxy for liquidity and sort the coins based on the liquidity. Panel A (B) reports the risk-adjusted alphas for high (low) liquidity sub-sample and the differences between risk-adjusted alphas for the sub-samples are reported in Panel C. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: High liquidity							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-0.82***	-0.85***	-0.83***	-1.13***	-1.13***	-1.11***	-1.02***
2	-0.25***	-0.28***	-0.26***	-0.56***	-0.55***	-0.52***	-0.44***
3	0.00	-0.03***	-0.02	-0.25***	-0.24***	-0.21***	-0.11***
4	0.26***	0.23***	0.24***	0.06	0.07*	0.11***	0.22***
Long	0.84***	0.81***	0.82***	0.70***	0.73***	0.82***	0.92***
Long - Short	1.65***	1.65***	1.65***	1.83***	1.86***	1.93***	1.94***
T-stats	*(60.77)*	*(58.10)*	*(55.25)*	*(53.72)*	*(57.46)*	*(60.93)*	*(51.39)*
Panel B: Low liquidity							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-2.19***	-2.22***	-2.2***	-2.17***	-2.26***	-2.34***	-2.04***
2	-0.40***	-0.42***	-0.41***	-0.66***	-0.67***	-0.66***	-0.55***
3	0.05***	0.02*	0.04**	-0.24***	-0.25***	-0.21***	-0.13***
4	0.54***	0.51***	0.53***	0.23***	0.25***	0.31***	0.37***
Long	2.54***	2.52***	2.54***	2.06***	2.18***	2.33***	2.20***
Long - Short	4.74***	4.74***	4.74***	4.23***	4.44***	4.66***	4.24***
T-stats	*(82.86)*	*(125.21)*	*(147.58)*	*(91.70)*	*(95.56)*	*(96.88)*	*(93.23)*
Panel C: Difference between high and low liquidity							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	1.37***	1.37***	1.37***	1.04***	1.14***	1.22***	1.02***
2	0.15***	0.15***	0.15***	0.10***	0.12***	0.14***	0.11***
3	-0.05***	-0.05***	-0.05***	-0.01	0.01	0.00	0.02
4	-0.28***	-0.28***	-0.29***	-0.18***	-0.18***	-0.20***	-0.14***
Long	-1.71***	-1.71***	-1.71***	-1.36***	-1.45***	-1.51***	-1.28***
Long - Short	-3.08***	-3.08***	-3.08***	-2.4***	-2.58***	-2.74***	-2.31***
T-stats	*(-79.79)*	*(-74.39)*	*(-68.91)*	*(-60.11)*	*(-62.36)*	*(-68.70)*	*(-52.04)*

Table XI Long-short portfolio performance for different market conditions

Table XI reports risk-adjusted alphas (in bps) for different market conditions using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors. We divide the whole sample into terciles based on the CRIX index proposed by Trimborn and Härdle (2018) and treat the top tercile as market boom and the rest as normal days. Panel A (B) reports the risk-adjusted alphas for market boom (normal) days and the differences between risk-adjusted alphas for the sub-samples are reported in Panel C. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Market boom days							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.59***	-1.62***	-1.60***	-1.70***	-1.76***	-1.81***	-1.59***
2	-0.28***	-0.31***	-0.30***	-0.58***	-0.58***	-0.56***	-0.47***
3	0.02	-0.01	0.00	-0.25***	-0.24***	-0.23***	-0.12***
4	0.32***	0.29***	0.30***	0.05	0.07	0.11*	0.19***
Long	1.71***	1.68***	1.69***	1.38***	1.45***	1.56***	1.55***
Long - Short	3.30***	3.30***	3.30***	3.09***	3.22***	3.37***	3.14***
T-stats	*(60.88)*	*(72.26)*	*(75.81)*	*(62.74)*	*(72.09)*	*(85.50)*	*(65.46)*
Panel B: Normal days							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.65***	-1.67***	-1.65***	-1.79***	-1.83***	-1.84***	-1.66***
2	-0.28***	-0.30***	-0.28***	-0.58***	-0.57***	-0.54***	-0.45***
3	0.01	-0.01	0.01	-0.25***	-0.26***	-0.20***	-0.11***
4	0.33***	0.31***	0.33***	0.07	0.09**	0.15***	0.22***
Long	1.78***	1.76***	1.78***	1.52***	1.58***	1.71***	1.71***
Long - Short	3.43***	3.43***	3.43***	3.31***	3.41***	3.55***	3.38***
T-stats	*(61.81)*	*(72.85)*	*(63.93)*	*(63.00)*	*(81.92)*	*(70.66)*	*(60.02)*
Panel C: Difference between market boom and normal days							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	0.06	0.05	0.05	0.06	0.05	0.05	0.05
2	0.00	-0.01	-0.02	-0.02	-0.03	-0.02	-0.03
3	0.01	-0.01	-0.01	-0.01	-0.02	-0.01	-0.01
4	-0.01	-0.02	-0.03	-0.02	-0.03	-0.02	-0.02
Long	-0.07*	-0.08*	-0.09*	-0.09**	-0.09**	-0.08*	-0.09**
Long - Short	-0.14	-0.14	-0.14	-0.15*	-0.14*	-0.13	-0.14*
T-stats	*(48.96)*	*(49.82)*	*(49.45)*	*(48.90)*	*(50.55)*	*(51.84)*	*(53.53)*

Table XII Long-short portfolio performance for different uncertainty periods

Table XII reports risk-adjusted alphas (in bps) for different uncertainty periods using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors. We follow Al-Shboul, Assaf, and Mokni (2022) and use the crypto policy uncertainty (CCPO) proposed by Lucey, Vigne, Yarovaya, and Wang (2021) as the market uncertainty proxy. We divide the whole sample into terciles based on the CCPO and identify days in the top tercile as high uncertainty days and the rest as low uncertainty days. Panel A (B) reports the risk-adjusted alphas for high (low) uncertainty periods and the differences between risk-adjusted alphas for the sub-samples are reported in Panel C. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: High uncertainty							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.49***	-1.60***	-1.53***	-1.44***	-1.47***	-1.38***	-1.53***
2	0.12	0.04	0.09	0.17	0.14	0.15	0.04
3	0.21	0.12	0.17	0.24	0.21	0.24	0.14
4	0.25	0.14	0.20	0.31	0.26	0.30	0.19
Long	2.17***	2.04***	2.10***	2.31***	2.22***	2.41***	2.27***
Long - Short	3.66***	3.64***	3.64***	3.75***	3.69***	3.79***	3.79***
T-stats	*(6.23)*	*(6.11)*	*(6.49)*	*(6.15)*	*(6.26)*	*(6.52)*	*(6.65)*
Panel B: Low uncertainty							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.61***	-1.64***	-1.62***	-1.74***	-1.79***	-1.82***	-1.62***
2	-0.28***	-0.31***	-0.29***	-0.58***	-0.58***	-0.55***	-0.46***
3	0.02	-0.01	0.00	-0.25***	-0.25***	-0.22***	-0.12***
4	0.32***	0.29***	0.31***	0.06	0.07*	0.12***	0.21***
Long	1.74***	1.71***	1.73***	1.43***	1.50***	1.62***	1.61***
Long - Short	3.35***	3.35***	3.35***	3.17***	3.29***	3.44***	3.23***
T-stats	*(93.32)*	*(84.56)*	*(86.51)*	*(81.82)*	*(91.67)*	*(92.91)*	*(81.06)*
Panel C: Difference between high and low uncertainty							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	0.12	-0.04	0.08	0.20	0.27	0.24	-0.07
2	0.40	0.24	0.36	0.60	0.66*	0.56	0.34
3	0.19	0.03	0.15	0.37	0.42	0.33	0.10
4	-0.08	-0.24	-0.12	0.09	0.14	0.04	-0.18
Long	0.43	0.27	0.39	0.63	0.64	0.50	0.34
Long - Short	0.31	0.31	0.31	0.43	0.36	0.26	0.41
T-stats	*(0.66)*	*(0.89)*	*(0.85)*	*(0.90)*	*(0.72)*	*(0.70)*	*(0.98)*

Table XIII Time-series regression analysis of alternative channels

Table XIII reports coefficient estimates of time-series regressions of the minute-level cryptocurrency return forecasts generated by the adaptive LASSO method on three alternative channel proxies. For liquidity, we use the return of the long-short portfolio of Amihud illiquidity constructed in Table IX. For uncertainty, we follow Al-Shboul, Assaf, and Mokni (2022) and use the crypto policy uncertainty (CCPO) proposed by Lucey, Vigne, Yarovaya, and Wang (2021). The dummy variable of market uncertainty (equals one if CCPO index is larger than the time-series median and zero otherwise) is included as the independent variable. For wealth effect, we use the CRIX index proposed by Trimborn and Härdle (2018) and use the dummy variable of market wealth (equals one if CRIX index is larger than the time-series median and zero otherwise) as the proxy. Column (1) – (3) present results of including one of the proxy in the regression and Column (4) reports results of including all three proxies. The Newey-West adjusted t-statistics are reported in the parentheses. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Intercept	0.03***	0.03***	0.03***	0.03***
	(57.94)	(57.41)	(19.19)	(19.05)
Liquidity	0.00			0.00
	(0.44)			(0.44)
Uncertainty		-0.00		-0.00
		(-0.29)		(-0.33)
Wealth Effect			0.00	0.00
			(1.18)	(1.24)

Table XIV Granger causality test results

Table XIV reports the results for Granger causality test. Panel A shows the ADF test results and Panel B shows the F-stats and p-values for the Granger causality test. We use daily Google search volume to proxy for investor attention. Another key variable is the cumulative daily returns of minute-level portfolio returns constructed based on the adaptive LASSO method. The sample period is from 25 March 2019 to 30 April 2021.

Panel A ADF tests		
	ADF-stat	p-value
Google search volume	-8.40	0.01
Portfolio return of adaptive LASSO	-56.16	0.01
Panel B Granger causality test		
Null hypothesis:	F-stat	p-value
Google search volume does not Granger-cause portfolio return of adaptive LASSO	11.15	0.00
Portfolio return of adaptive LASSO does not Granger-cause Google search volume	0.01	0.92

Table XV Long-short portfolio performance for different correlation periods

Table XV reports risk-adjusted alphas (in bps) for different correlation periods using minute-level cryptocurrency return forecasts generated by the adaptive LASSO method. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors. For each half-day rolling window, we compute average correlation following Pollet and Wilson (2010). We divide the whole sample into terciles based on the average correlation and treat the top tercile as high correlation periods and the rest as low correlation periods. Panel A (B) reports the risk-adjusted alphas for high (low) correlation periods and the differences between risk-adjusted alphas for the sub-samples are reported in Panel C. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: High correlation							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-2.35***	-1.95***	-2.17***	-3.04***	-3.10***	-3.16***	-1.97***
2	-0.98***	-0.55***	-0.78***	-1.85***	-1.84***	-1.89***	-0.70***
3	-0.52**	-0.13	-0.33*	-1.38***	-1.41***	-1.41***	-0.37**
4	-0.14	0.29**	0.07	-1.18**	-1.23**	-1.17***	-0.09
Long	1.33***	1.71***	1.51***	0.45	0.46	0.51	1.44***
Long - Short	3.68***	3.66***	3.68***	3.49***	3.57***	3.67***	3.41***
T-stats	*(11.19)*	*(7.97)*	*(9.05)*	*(12.48)*	*(12.86)*	*(10.6)*	*(14.78)*
Panel B: Low correlation							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.60***	-1.63***	-1.61***	-1.73***	-1.78***	-1.81***	-1.61***
2	-0.27***	-0.30***	-0.28***	-0.57***	-0.57***	-0.54***	-0.46***
3	0.02*	-0.01	0.01	-0.24***	-0.24***	-0.21***	-0.12***
4	0.33***	0.29***	0.32***	0.06	0.08**	0.13***	0.20***
Long	1.74***	1.71***	1.73***	1.44***	1.51***	1.62***	1.61***
Long - Short	3.34***	3.34***	3.34***	3.17***	3.29***	3.43***	3.22***
T-stats	*(72.22)*	*(86.66)*	*(81.69)*	*(80.16)*	*(89.48)*	*(91.59)*	*(81.62)*
Panel C: Difference between high and low correlation							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-0.75	-0.31	-0.61	-0.75	-0.75	-0.74	-0.29
2	-0.71**	-0.26	-0.56*	-0.71*	-0.71*	-0.70*	-0.25
3	-0.54**	-0.08	-0.38**	-0.54**	-0.54**	-0.53**	-0.08
4	-0.47**	-0.01	-0.32**	-0.47**	-0.47**	-0.46**	-0.01
Long	-0.41**	0.04	-0.27	-0.41**	-0.41**	-0.40**	0.05
Long - Short	0.34	0.35	0.34	0.34	0.34	0.34	0.34
T-stats	*(0.56)*	*(0.67)*	*(0.63)*	*(0.72)*	*(0.71)*	*(0.64)*	*(0.72)*

Table XVI Portfolio performance and trading costs

Table XVI presents the economic gains of portfolios with and without trading costs in the spot and futures markets. For each ten minute, we sort the 30 sample coins in ascending order based on the excess return forecasts generated from the adaptive LASSO technique using half-day and one-day rolling windows and group them into five equal-weighted portfolios with a ten-minute rebalancing frequency. We subsequently construct a long-short portfolio that goes long (short) the top (bottom) quintile portfolio. We use corresponding trading fee levels of takers for the futures (4 bps) according to *Binance* for its regular users. For the sport market, we follow Jia, Wu, Yan, and Liu (2023) to use 12 bps as the total transaction cost, including bid-ask spread and trading fees. Ten-minute returns in bps are reported. The sample period for spot is 25 March 2019 to 30 April 2021 and for futures is 29 July 2020 to 30 April 2021. The t-statistics are Newey-West adjusted. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Futures		**Spot**	
	Half-day Window	One-day Window	Half-day Window	One-day Window
Short	0.18	-0.07	-1.69***	-2.25***
2	0.66***	0.47**	0.03	-0.35***
3	0.61***	0.55***	0.28**	0.33***
4	0.73***	1.03***	0.61***	0.81***
Long	1.57***	1.74***	2.63***	3.32***
Long - Short (without Trading Costs)	1.39***	1.80***	4.31***	5.58***
Long - Short (with Trading Costs)	0.34***	0.66***	0.74***	1.83***

Figure 1 Loadings on the first three principal components

Figure 1 plots the loadings on the first three components extracted from the cross-cryptocurrency portfolio excess returns. Factor loadings of the first three principal components, which are extracted from the minute-level standardized returns of all 30 cryptocurrencies, are shown in the figure. The sample period is from 25 March 2019 to 30 April 2021.

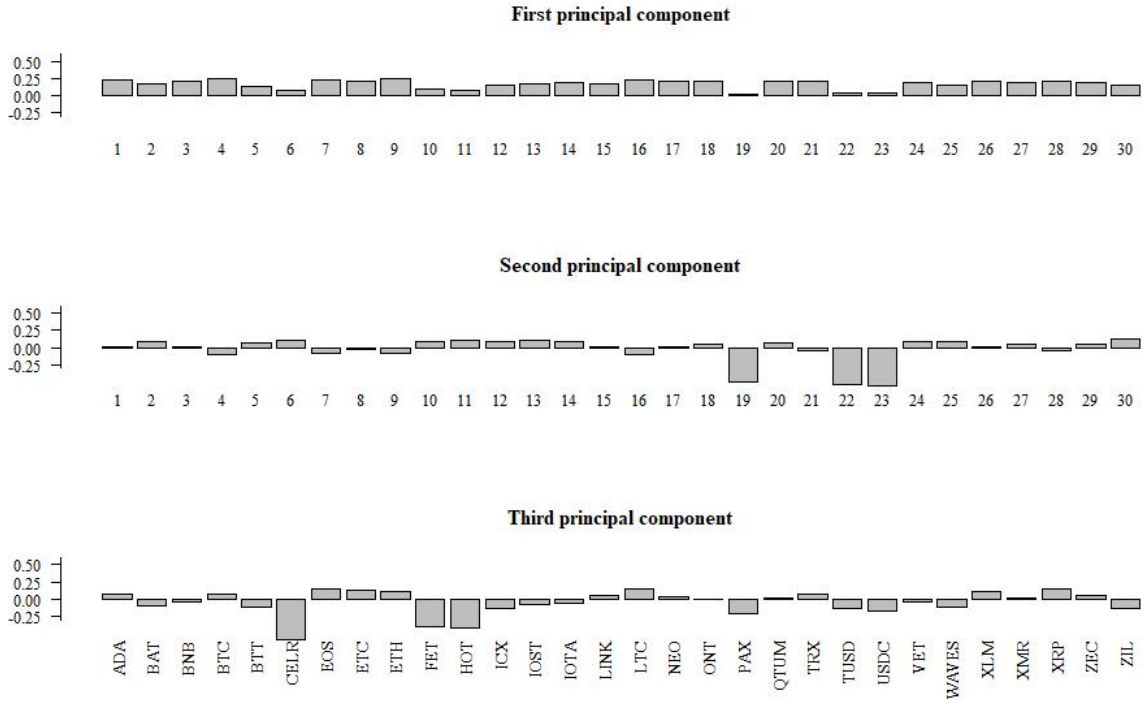


Figure 2 Time-series weight of the first principal component in PCA

Figure 2 plots time-series weights of the first component for top three coins (BTC, ETH, LTC) and bottom three coins (PAX, TUSD, USDC) in PCA. The sample period is from 25 March 2019 to 30 April 2021.

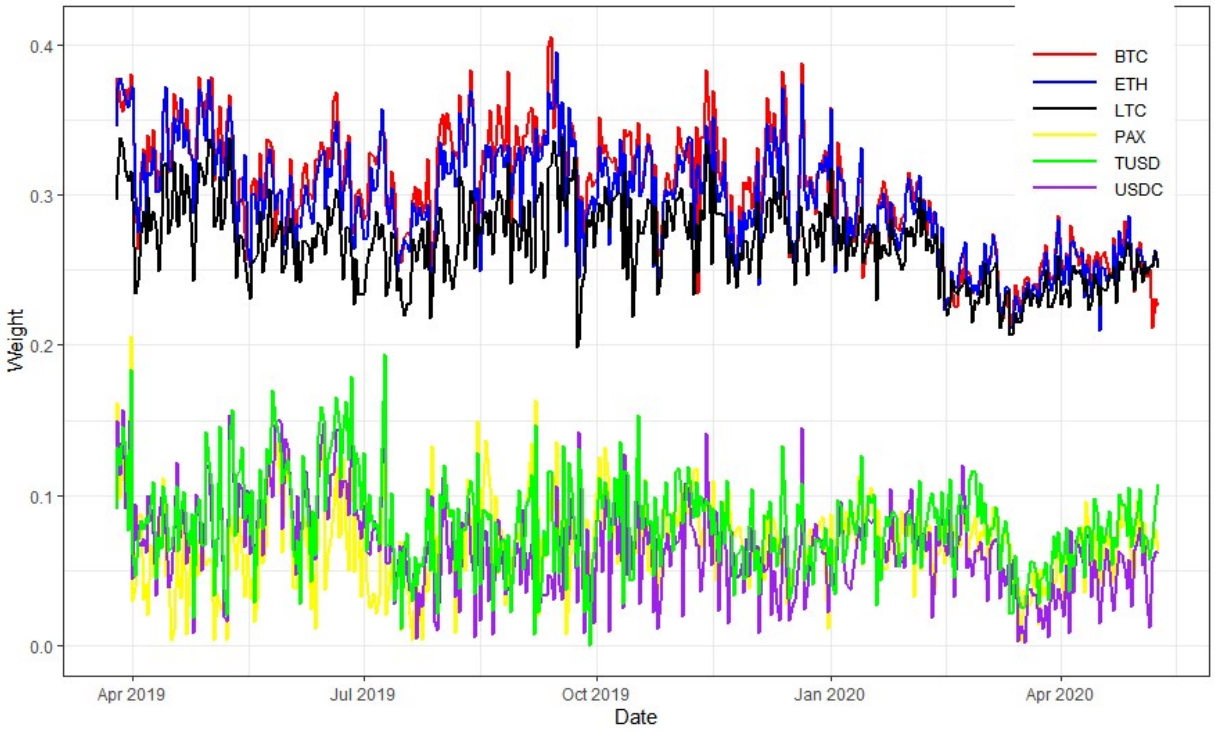
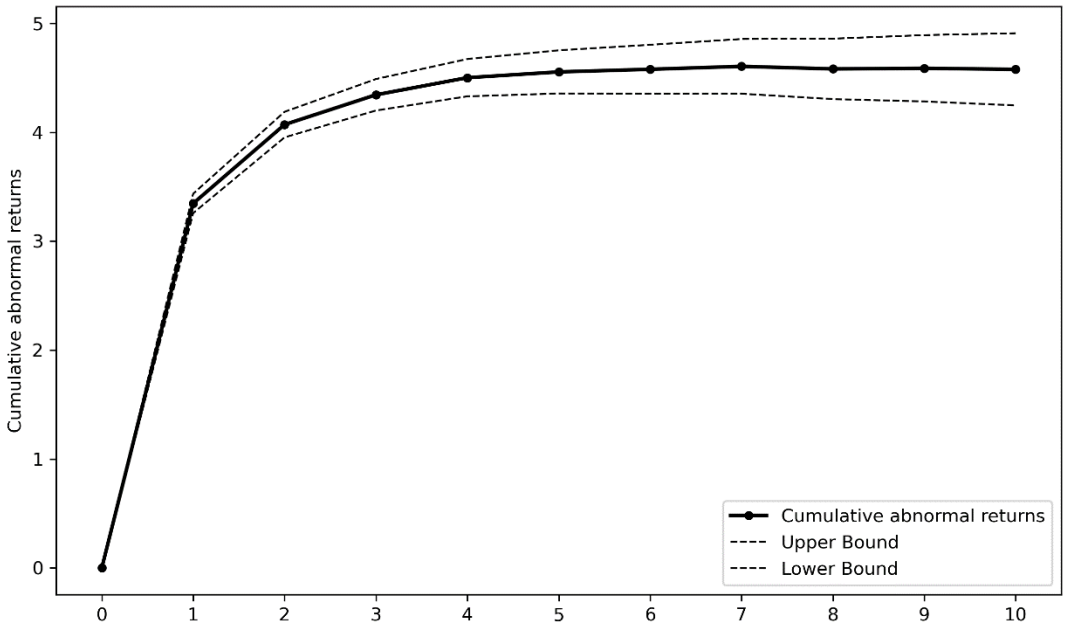


Figure 3 Cumulative excess return over the time horizon

Figure 3 shows the cumulative excess returns of the long-short portfolio predicted by the adaptive LASSO up to the next ten minutes (x-axis) with the corresponding 95% confidence intervals (shown as the upper and lower bounds). Returns are minute-level returns in bps. The sample period is from 25 March 2019 to 30 April 2021.



Appendix of "Cross-Cryptocurrency Return Predictability"

Table A.I Univariate OLS predictive regression results

Table A.I reports coefficient estimates (in bps) for each of the 30 cryptocurrencies from the univariate OLS regressions. The minute-level returns are standardized to have mean of 0 and unit variance. Return of each cryptocurrency in the row serves as the dependent variable and lagged return of each cryptocurrency in the column is the independent variable. Each coefficient estimate corresponds to one univariate OLS regression model. For each cryptocurrency, the average value of their coefficient estimates for other coins (estimates for stablecoins are excluded) is reported as Avg_Coef. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Avg_Coef	ADA	BAT	BNB	BTC	BTT	CELR
ADA	1.09	-0.04*	1.44***	0.47***	0.06***	1.42***	2.41***
BAT	0.64	0.38***	-1.27***	0.15***	-0.06***	0.87***	1.75***
BNB	1.13	0.77***	1.47***	-0.20***	0.07***	1.39***	2.48***
BTC	1.39	0.93***	1.62***	0.63***	-0.02	1.81***	2.83***
BTT	0.56	0.35***	0.67***	0.25***	-0.03*	-0.56***	1.48***
CELR	0.25	0.15***	0.28***	0.05***	-0.03*	0.39***	-2.84***
EOS	1.23	0.89***	1.56***	0.59***	0.02*	1.64***	2.44***
ETC	1.03	0.70***	1.36***	0.49***	0.03**	1.42***	2.24***
ETH	1.36	0.97***	1.75***	0.64***	0.05***	1.71***	2.66***
FET	0.41	0.23***	0.51***	0.10***	-0.03**	0.64***	1.46***
HOT	0.23	0.05**	0.26***	0.03*	-0.09***	0.40***	0.86***
ICX	0.63	0.39***	0.85***	0.25***	0.01	0.84***	1.76***
IOST	0.66	0.43***	0.86***	0.27***	-0.02	1.04***	1.84***
IOTA	0.71	0.42***	0.98***	0.26***	-0.03**	1.03***	1.93***
LINK	0.91	0.63***	1.23***	0.42***	0.04***	1.22***	1.96***
LTC	1.27	0.96***	1.61***	0.59***	0.11***	1.63***	2.52***
NEO	1.05	0.73***	1.35***	0.46***	0.08***	1.46***	2.27***
ONT	0.99	0.63***	1.22***	0.43***	0.03**	1.44***	2.37***
PAX	0.15	0.14***	0.25***	0.02	-0.09***	0.25***	0.23***
QTUM	0.88	0.57***	1.11***	0.37***	0.01	1.28***	2.12***
TRX	1.14	0.86***	1.43***	0.56***	0.09***	1.88***	2.34***
TUSD	0.31	0.19***	0.47***	0.10***	0.06***	0.37***	0.22***
USDC	0.21	0.14***	0.40***	0.04**	-0.03**	0.29***	0.26***
VET	0.77	0.45***	1.02***	0.23***	-0.03**	1.06***	2.04***
WAVES	0.57	0.32***	0.66***	0.20***	-0.01	0.82***	1.51***
XLM	1.02	0.86***	1.33***	0.40***	0.03**	1.27***	2.29***
XMR	0.80	0.52***	1.09***	0.34***	0.01	1.10***	1.97***
XRP	0.97	0.67***	1.17***	0.42***	0.01	1.27***	2.17***
ZEC	0.98	0.74***	1.24***	0.49***	0.12***	1.34***	2.13***
ZIL	0.55	0.29***	0.67***	0.17***	-0.03*	0.84***	1.64***

Table A.I (Continued) Univariate OLS predictive regression results

	EOS	ETC	ETH	FET	HOT	ICX
ADA	0.32***	0.64***	0.18***	2.53***	2.28***	1.60***
BAT	0.18***	0.33***	0.02	1.69***	1.57***	1.00***
BNB	0.39***	0.64***	0.15***	2.60***	2.48***	1.62***
BTC	0.38***	0.89***	0.23***	2.89***	2.86***	1.78***
BTT	0.18***	0.23***	0.02	1.46***	1.53***	0.79***
CELR	0.08***	0.12***	-0.03*	1.00***	0.79***	0.36***
EOS	-0.06***	0.83***	0.19***	2.69***	2.39***	1.70***
ETC	0.32***	-0.24***	0.13***	2.39***	2.13***	1.49***
ETH	0.30***	0.79***	-0.07***	2.98***	2.78***	1.83***
FET	0.14***	0.23***	0.04**	-2.86***	1.10***	0.62***
HOT	0.05**	0.10***	-0.05***	0.78***	-2.64***	0.33***
ICX	0.17***	0.31***	0.08***	1.76***	1.46***	-1.91***
IOST	0.15***	0.31***	0.04**	1.78***	1.66***	1.03***
IOTA	0.19***	0.36***	0.06***	1.94***	1.75***	1.07***
LINK	0.25***	0.51***	0.14***	2.10***	1.83***	1.31***
LTC	0.34***	0.79***	0.22***	2.74***	2.41***	1.77***
NEO	0.37***	0.62***	0.20***	2.48***	2.15***	1.45***
ONT	0.38***	0.62***	0.14***	2.36***	2.12***	1.43***
PAX	-0.01	0.13***	-0.01	0.17***	0.25***	0.19***
QTUM	0.28***	0.54***	0.13***	2.13***	1.89***	1.28***
TRX	0.36***	0.67***	0.22***	2.53***	2.13***	1.58***
TUSD	0.15***	0.17***	0.04***	0.42***	0.41***	0.48***
USDC	0.06***	0.12***	-0.11***	0.21***	0.27***	0.28***
VET	0.22***	0.43***	0.04**	2.00***	1.84***	1.14***
WAVES	0.14***	0.32***	0.08***	1.51***	1.35***	0.90***
XLM	0.36***	0.62***	0.12***	2.38***	1.99***	1.48***
XMR	0.18***	0.40***	0.12***	1.99***	1.92***	1.17***
XRP	0.30***	0.61***	0.10***	2.30***	1.86***	1.41***
ZEC	0.38***	0.59***	0.25***	2.22***	2.01***	1.39***
ZIL	0.17***	0.35***	0.05***	1.55***	1.33***	0.84***

Table A.I (Continued) Univariate OLS predictive regression results

	IOST	IOTA	LINK	LTC	NEO	ONT
ADA	1.77***	1.51***	0.76***	0.25***	0.95***	1.27***
BAT	1.08***	0.85***	0.43***	0.12***	0.47***	0.74***
BNB	1.73***	1.51***	0.90***	0.28***	0.99***	1.33***
BTC	2.35***	1.85***	1.07***	0.30***	1.12***	1.60***
BTT	0.92***	0.79***	0.36***	0.08***	0.46***	0.72***
CELR	0.47***	0.30***	0.10***	0.07***	0.18***	0.30***
EOS	1.95***	1.66***	0.96***	0.23***	1.08***	1.52***
ETC	1.62***	1.43***	0.73***	0.17***	0.88***	1.29***
ETH	2.22***	1.84***	1.06***	0.24***	1.18***	1.59***
FET	0.67***	0.53***	0.25***	0.09***	0.31***	0.45***
HOT	0.39***	0.33***	0.09***	-0.02	0.19***	0.28***
ICX	1.08***	0.86***	0.38***	0.15***	0.44***	0.73***
IOST	-0.13***	0.95***	0.36***	0.10***	0.56***	0.86***
IOTA	1.16***	-1.23***	0.44***	0.13***	0.53***	0.87***
LINK	1.48***	1.31***	-1.06***	0.20***	0.73***	1.09***
LTC	2.00***	1.66***	1.11***	-0.21***	1.13***	1.53***
NEO	1.70***	1.45***	0.70***	0.27***	-0.32***	1.37***
ONT	1.75***	1.33***	0.63***	0.23***	0.87***	0.21***
PAX	0.29***	0.17***	0.20***	0.01	0.15***	0.20***
QTUM	1.47***	1.19***	0.61***	0.20***	0.83***	1.21***
TRX	1.78***	1.49***	0.91***	0.24***	0.98***	1.36***
TUSD	0.61***	0.40***	0.28***	0.06***	0.31***	0.36***
USDC	0.44***	0.33***	0.27***	-0.01	0.25***	0.32***
VET	1.30***	1.09***	0.41***	0.12***	0.64***	0.94***
WAVES	0.96***	0.79***	0.33***	0.10***	0.44***	0.65***
XLM	1.62***	1.40***	0.64***	0.24***	0.91***	1.18***
XMR	1.37***	1.10***	0.52***	0.12***	0.58***	0.89***
XRP	1.47***	1.35***	0.69***	0.18***	0.90***	1.17***
ZEC	1.57***	1.35***	0.69***	0.31***	0.86***	1.25***
ZIL	0.93***	0.72***	0.30***	0.14***	0.44***	0.66***

Table A.I (Continued) Univariate OLS predictive regression results

	PAX	QTUM	TRX	TUSD	USDC	VET
ADA	-0.10***	1.24***	0.72***	-0.16***	-0.15***	1.18***
BAT	-0.06***	0.66***	0.38***	-0.08***	-0.09***	0.76***
BNB	-0.09***	1.21***	0.69***	-0.12***	-0.14***	1.22***
BTC	-0.16***	1.62***	0.74***	-0.23***	-0.24***	1.59***
BTT	-0.06***	0.62***	0.54***	-0.09***	-0.08***	0.60***
CELR	-0.03***	0.28***	0.16***	-0.05***	-0.05***	0.26***
EOS	-0.12***	1.44***	0.78***	-0.18***	-0.18***	1.29***
ETC	-0.09***	1.19***	0.62***	-0.14***	-0.14***	1.09***
ETH	-0.14***	1.55***	0.78***	-0.19***	-0.20***	1.46***
FET	-0.05***	0.48***	0.23***	-0.06***	-0.06***	0.45***
HOT	-0.04***	0.31***	0.13***	-0.05***	-0.04***	0.28***
ICX	-0.06***	0.70***	0.40***	-0.09***	-0.08***	0.71***
IOST	-0.07***	0.75***	0.43***	-0.10***	-0.10***	0.73***
IOTA	-0.08***	0.82***	0.43***	-0.11***	-0.12***	0.76***
LINK	-0.07***	0.96***	0.54***	-0.12***	-0.12***	1.05***
LTC	-0.12***	1.47***	0.79***	-0.18***	-0.18***	1.39***
NEO	-0.11***	1.29***	0.64***	-0.16***	-0.15***	1.22***
ONT	-0.10***	1.21***	0.61***	-0.13***	-0.14***	1.10***
PAX	-0.57***	0.28***	0.04**	0.00*	-0.04***	0.09***
QTUM	-0.09***	-0.34***	0.51***	-0.13***	-0.13***	0.98***
TRX	-0.11***	1.32***	-0.04**	-0.17***	-0.16***	1.23***
TUSD	-0.02***	0.44***	0.19***	-0.61***	-0.07***	0.32***
USDC	-0.02***	0.40***	0.12***	-0.05***	-0.76***	0.16***
VET	-0.06***	0.89***	0.46***	-0.10***	-0.09***	-0.89***
WAVES	-0.06***	0.60***	0.33***	-0.09***	-0.09***	0.62***
XLM	-0.09***	1.20***	0.64***	-0.13***	-0.13***	1.10***
XMR	-0.10***	0.91***	0.40***	-0.14***	-0.14***	0.88***
XRP	-0.09***	1.10***	0.55***	-0.13***	-0.12***	0.99***
ZEC	-0.09***	1.07***	0.56***	-0.13***	-0.12***	1.07***
ZIL	-0.05***	0.63***	0.38***	-0.07***	-0.08***	0.59***

Table A.I (Continued) Univariate OLS predictive regression results

	WAVES	XLM	XMR	XRP	ZEC	ZIL
ADA	1.31***	0.76***	1.08***	0.33***	1.11***	1.81***
BAT	0.89***	0.39***	0.60***	0.19***	0.63***	1.16***
BNB	1.45***	0.77***	1.03***	0.43***	1.06***	1.93***
BTC	1.79***	0.91***	1.55***	0.45***	1.53***	2.33***
BTT	0.67***	0.32***	0.47***	0.18***	0.57***	0.95***
CELR	0.32***	0.13***	0.20***	0.05**	0.25***	0.42***
EOS	1.53***	0.89***	1.24***	0.41***	1.37***	1.93***
ETC	1.28***	0.67***	1.06***	0.27***	1.19***	1.65***
ETH	1.72***	0.97***	1.43***	0.44***	1.51***	2.17***
FET	0.54***	0.25***	0.32***	0.12***	0.41***	0.83***
HOT	0.31***	0.13***	0.23***	0.04*	0.21***	0.52***
ICX	0.82***	0.37***	0.59***	0.17***	0.64***	1.16***
IOST	0.82***	0.43***	0.57***	0.18***	0.66***	1.16***
IOTA	0.88***	0.44***	0.66***	0.18***	0.74***	1.29***
LINK	1.19***	0.61***	0.87***	0.35***	0.94***	1.54***
LTC	1.54***	0.94***	1.31***	0.40***	1.40***	2.04***
NEO	1.24***	0.71***	1.05***	0.33***	1.17***	1.69***
ONT	1.26***	0.62***	0.94***	0.37***	1.09***	1.64***
PAX	0.26***	0.10***	0.20***	-0.01	0.28***	0.24***
QTUM	1.07***	0.53***	0.87***	0.27***	0.97***	1.43***
TRX	1.40***	0.82***	1.14***	0.36***	1.25***	1.79***
TUSD	0.56***	0.22***	0.39***	0.11***	0.52***	0.46***
USDC	0.38***	0.19***	0.31***	0.02	0.35***	0.32***
VET	0.98***	0.47***	0.70***	0.24***	0.76***	1.27***
WAVES	-1.72***	0.36***	0.53***	0.17***	0.57***	1.04***
XLM	1.24***	-0.04**	0.97***	0.43***	1.07***	1.67***
XMR	0.97***	0.50***	-0.43***	0.23***	0.85***	1.44***
XRP	1.18***	0.93***	0.95***	-0.05**	1.03***	1.54***
ZEC	1.16***	0.68***	1.01***	0.35***	0.14***	1.58***
ZIL	0.66***	0.34***	0.47***	0.18***	0.53***	-1.62***

Table A.II OLS predictive regression results: lagged BTC returns using 5-minute returns

Table A.II reports coefficient estimates for each of the 30 cryptocurrencies based on 5-minute returns, with the lagged return of Bitcoin as the independent variable and its own lagged return as control variable. Return of each cryptocurrency shown in the table serves as the dependent variable in the OLS regression. Each value in the row (*btc_lag*) is the coefficient estimate of the lagged return of Bitcoin. Returns are on a 5-minute basis and are standardized. Estimates are presented in bps. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	ADA	BAT	BNB	BTC	BTT	CELR
btc_lag	4.94***	8.68***	2.03***	–	10.85***	23.15***
	EOS	ETC	ETH	FET	HOT	ICX
btc_lag	2.70***	3.92***	0.49	14.27***	21.07***	8.74***
	IOST	IOTA	LINK	LTC	NEO	ONT
btc_lag	11.02***	10.35***	6.58***	4.23***	5.94***	8.78***
	PAX	QTUM	TRX	TUSD	USDC	VET
btc_lag	-1.10***	9.26***	5.20***	-2.15***	-1.40***	9.22***
	WAVES	XLM	XMR	XRP	ZEC	ZIL
btc_lag	10.18***	3.69***	11.18***	1.02**	9.61***	16.28***

Table A.III OLS predictive regression results using Bitfinex data: lagged BTC returns

Table A.III reports coefficient estimates for each of the cryptocurrencies, with the lagged return of Bitcoin as the independent variable and its own lagged return as control variable. We use 22 coins from Bitfinex following Jia, Wu, Yan, and Liu (2023). Return of each cryptocurrency shown in the table serves as the dependent variable in the OLS regression. Each value in the row (*btc_lag*) is the coefficient estimate of the lagged return of Bitcoin. Returns are on a five-minute basis and are standardized. Estimates are presented in bps. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	AID	AVT	BTC	BTG	DAT	EOS
btc_lag	0.43*	0.12	–	8.29***	4.59***	2.62***
	ETH	ETP	LTC	NEO	OMG	QSH
btc_lag	1.04***	4.88***	2.10***	4.78***	5.50***	0.99***
	QTM	SAN	SPK	TNB	TRX	XLM
btc_lag	7.30***	3.18***	0.60***	1.88***	5.74***	5.57***
	XMR	XRP	ZEC	ZRX		
btc_lag	5.15***	0.57***	5.24***	5.33***		

Table A.IV Adaptive LASSO predictive regression results

Table A.IV reports coefficient estimates for each of the 30 cryptocurrencies from the adaptive LASSO predictive model. Return of each cryptocurrency in the top header serves as the dependent variable in the

adaptive LASSO model. Lagged return of each cryptocurrency in the left column serves as the independent variable. − indicates that the corresponding cryptocurrency in the left column was not selected as a return predictor by the adaptive LASSO. Estimates are presented in bps and those in bold are statistically significant based on their bootstrapped 90% confidence intervals. The sample period is from 25 March 2019 to 30 April 2021.

	ADA	BAT	BNB	BTC	BTT	CELR
ADA	**-8.79**	**1.83**	**0.68**	0.25	**0.23**	**1.15**
BAT	**-0.43**	**-14.35**	**-0.69**	**-0.49**	**-0.49**	**0.74**
BNB	**1.79**	**3.87**	**-6.59**	**0.55**	**1.59**	**4.53**
BTC	**2.08**	**1.20**	**3.55**	**-1.89**	**5.56**	**8.19**
BTT	**-0.21**	−	0.20	-0.22	**-8.36**	**0.85**
CELR	-0.01	−	-0.07	−	**0.05**	**-12.35**
EOS	**1.44**	**1.56**	**1.23**	-0.41	**1.40**	**0.16**
ETC	−	**0.95**	**0.55**	−	**0.85**	**1.06**
ETH	**3.22**	**4.70**	**2.11**	**0.32**	**-0.30**	**-1.38**
FET	-0.07	−	-0.15	-0.01	**0.28**	**2.17**
HOT	**-0.60**	-0.33	**-0.26**	**-0.33**	**-0.08**	**0.49**
ICX	−	**0.58**	−	−	**-0.15**	**1.58**
IOST	−	−	−	-0.15	**0.49**	**1.03**
IOTA	**-0.59**	**0.34**	**-0.34**	-0.32	**-0.30**	**0.69**
LINK	**0.63**	**1.51**	**0.53**	−	**0.80**	**0.96**
LTC	**2.42**	**1.39**	**0.82**	**1.20**	**0.58**	**0.42**
NEO	**0.94**	**1.52**	**0.52**	**0.53**	**1.22**	**0.58**
ONT	**0.21**	**0.76**	**0.51**	−	**1.48**	**2.20**
PAX	−	−	**-2.49**	**-4.47**	−	**-2.36**
QTUM	−	0.36	−	−	**0.77**	**1.02**
TRX	**2.31**	**2.09**	**1.51**	**0.62**	**8.02**	**2.00**
TUSD	−	**3.21**	−	**2.70**	**1.24**	**-9.23**
USDC	**-0.04**	**1.79**	**-2.18**	-1.15	**-0.56**	**-2.80**
VET	**-0.19**	0.50	**-0.46**	-0.34	**-0.02**	**1.39**
WAVES	**-0.31**	−	-0.15	−	**0.02**	**0.49**
XLM	**3.38**	**2.03**	−	−	**-0.06**	**1.43**
XMR	**-0.62**	0.15	-0.21	−	**-0.80**	−
XRP	−	-0.31	−	-0.17	**-0.29**	**0.83**
ZEC	**1.48**	**1.50**	**1.17**	**0.85**	**1.27**	**0.90**
ZIL	**-0.36**	−	-0.23	-0.15	**0.28**	**1.28**

Table A.IV (Continued) Adaptive LASSO predictive regression results

	EOS	ETC	ETH	FET	HOT	ICX
ADA	**0.17**	**0.51**	**0.70**	**1.55**	**1.39**	**2.12**
BAT	**-0.19**	**-0.43**	**-0.49**	**0.45**	**0.51**	**0.71**

BNB	**1.36**	**1.02**	**0.38**	**5.27**	**5.69**	**3.96**
BTC	**2.09**	**4.55**	**4.02**	**6.50**	**8.72**	**3.54**
BTT	–	**-0.55**	**-0.42**	**0.62**	**2.02**	–
CELR	-0.01	**-0.08**	**-0.23**	**1.62**	**0.76**	–
EOS	**-6.06**	**3.53**	**0.74**	**1.16**	**1.36**	**1.40**
ETC	**0.96**	**-8.65**	–	**1.12**	**1.03**	**1.04**
ETH	**-0.81**	**0.93**	**-7.04**	**3.98**	**5.27**	**2.66**
FET	–	**-0.02**	**-0.03**	**-15.23**	**0.77**	**0.33**
HOT	**-0.17**	**-0.25**	**-0.28**	**0.20**	**-12.42**	**-0.27**
ICX	-0.11	**-0.27**	–	**1.57**	**0.31**	**-15.75**
IOST	**-0.43**	**-0.49**	**-0.36**	**0.67**	**0.85**	**0.50**
IOTA	**-0.37**	**-0.61**	**-0.27**	**0.73**	**0.34**	**0.22**
LINK	0.04	**0.30**	**0.38**	**1.47**	**0.53**	**1.22**
LTC	**1.40**	**2.01**	**1.71**	**1.02**	**-0.59**	**1.89**
NEO	**0.97**	**0.66**	**0.91**	**2.03**	**0.76**	**1.27**
ONT	**1.24**	**1.08**	–	**1.27**	**1.34**	**1.50**
PAX	**-2.55**	–	**-0.69**	**-7.47**	**-2.18**	**-3.24**
QTUM	**0.23**	**0.49**	0.11	**0.39**	**0.22**	**0.77**
TRX	**1.33**	**1.42**	**1.17**	**2.69**	**0.48**	**2.20**
TUSD	**4.64**	–	**1.56**	–	**-1.37**	**5.43**
USDC	–	**-1.06**	**-5.79**	**-9.32**	**-7.23**	**-0.05**
VET	-0.09	–	**-0.41**	**0.90**	**0.97**	**0.47**
WAVES	**-0.24**	**-0.11**	–	**0.76**	**0.23**	**0.77**
XLM	**0.92**	**0.76**	–	**1.59**	–	**1.89**
XMR	**-0.81**	**-0.99**	–	–	**0.44**	–
XRP	**0.59**	**0.51**	**-0.21**	**0.81**	**-0.54**	**0.33**
ZEC	**1.25**	**1.13**	**1.24**	**1.14**	**0.97**	**1.51**
ZIL	–	**0.11**	**-0.07**	**0.82**	**0.37**	**0.47**

Table A.IV (Continued) Adaptive LASSO predictive regression results

	IOST	IOTA	LINK	LTC	NEO	ONT
ADA	**1.37**	**2.09**	**0.93**	**0.65**	**1.39**	**0.82**
BAT	**-0.13**	0.05	–	**-0.12**	**-0.48**	**-0.38**
BNB	**2.07**	**3.17**	**3.12**	**1.15**	**2.49**	**2.42**
BTC	**10.12**	**6.23**	**4.80**	**3.59**	**3.17**	**3.39**
BTT	–	**0.31**	**-0.17**	**-0.23**	**-0.09**	**0.20**
CELR	**0.02**	-0.08	**-0.23**	–	**-0.13**	**-0.09**
EOS	**1.56**	**1.97**	**1.40**	**1.45**	**1.85**	**2.85**
ETC	**0.30**	**1.23**	–	**0.15**	**0.59**	**1.21**
ETH	**2.89**	**3.33**	**4.11**	**1.20**	**3.38**	**1.84**
FET	**-0.04**	–	–	–	**-0.14**	**-0.19**
HOT	**-0.41**	-0.17	**-0.44**	**-0.32**	**-0.24**	**-0.36**
ICX	**0.20**	**0.44**	-0.06	–	**-0.32**	**-0.06**
IOST	**-8.24**	**0.42**	**-0.55**	**-0.25**	**0.04**	**0.38**
IOTA	**-0.80**	**-17.83**	**-0.56**	**-0.28**	**-0.61**	**-0.31**
LINK	**0.94**	**1.45**	**-10.48**	**0.17**	**0.29**	**0.90**
LTC	**0.73**	**0.48**	**3.62**	**-7.23**	**2.01**	**1.70**
NEO	**1.13**	**2.19**	**0.26**	**0.90**	**-12.72**	**3.20**
ONT	**2.82**	**1.39**	–	**0.30**	**2.22**	**-9.54**
PAX	**-1.71**	**-4.61**	**0.93**	–	**-1.60**	**-0.99**
QTUM	**0.69**	**0.69**	**0.52**	**0.04**	**1.97**	**2.33**
TRX	**2.03**	**1.97**	**2.43**	**0.81**	**1.94**	**2.30**
TUSD	**7.23**	–	**1.41**	**0.50**	**4.22**	**2.37**
USDC	–	–	**2.12**	**-1.52**	**2.52**	**2.11**
VET	**0.36**	**0.70**	**-0.54**	**-0.28**	**0.07**	**0.20**
WAVES	–	0.23	**-0.14**	**-0.12**	**-0.16**	**-0.31**
XLM	**1.16**	**1.85**	**0.26**	**0.58**	**1.41**	**0.75**
XMR	**-0.41**	–	**-0.54**	**-0.63**	**-1.14**	**-1.50**
XRP	**-0.65**	**0.75**	–	–	**0.71**	–
ZEC	**1.42**	**1.88**	**1.23**	**1.39**	**1.56**	**2.11**
ZIL	–	–	**-0.22**	**0.01**	**-0.07**	**-0.04**

Table A.IV (Continued) Adaptive LASSO predictive regression results

	PAX	QTUM	TRX	TUSD	USDC	VET
ADA	–	**1.01**	**1.46**	**-0.08**	–	**1.34**
BAT	0.01	**-0.51**	**-0.24**	**0.06**	**0.02**	**0.40**
BNB	0.09	**1.36**	**1.47**	**0.24**	**0.12**	**2.52**
BTC	**-0.77**	**5.90**	**1.35**	**-1.00**	**-1.06**	**8.03**
BTT	–	**-0.05**	**1.81**	**-0.04**	**-0.05**	–
CELR	–	**-0.08**	–	**-0.05**	**-0.03**	-0.08
EOS	–	**2.29**	**2.50**	**-0.14**	**-0.22**	**0.64**
ETC	–	**0.95**	**0.63**	**0.04**	–	**0.38**
ETH	**-0.21**	**1.98**	**1.39**	**0.10**	**0.12**	**1.53**
FET	–	–	**-0.11**	**0.02**	–	–
HOT	-0.01	**-0.13**	**-0.24**	**-0.04**	**-0.01**	**-0.13**
ICX	–	–	**0.07**	0.01	**0.07**	**0.29**
IOST	–	–	**0.03**	**-0.05**	**-0.05**	**0.26**
IOTA	–	**-0.26**	**-0.27**	–	**-0.10**	**-0.33**
LINK	0.06	**0.33**	**0.42**	**0.00**	**0.01**	**1.30**
LTC	0.08	**1.41**	**1.82**	**0.07**	**0.11**	**1.58**
NEO	**-0.10**	**3.03**	**0.82**	**-0.10**	**0.03**	**2.15**
ONT	-0.01	**2.72**	**0.67**	**0.02**	**-0.07**	**1.42**
PAX	**-26.95**	**0.33**	**-2.61**	**5.18**	**4.85**	**-5.52**
QTUM	–	**-12.04**	–	**-0.08**	**-0.10**	**0.70**
TRX	-0.01	**2.38**	**-7.48**	**-0.20**	**-0.10**	**1.85**
TUSD	**4.02**	**3.89**	**3.38**	**-26.48**	**4.97**	**1.38**
USDC	**3.72**	**3.20**	–	**3.91**	**-32.23**	**-4.73**
VET	0.07	**0.27**	**-0.11**	**0.04**	**0.09**	**-11.96**
WAVES	–	**-0.27**	**-0.19**	**-0.02**	**-0.03**	–
XLM	–	**1.81**	**0.90**	–	0.02	**1.39**
XMR	**-0.16**	**-0.55**	**-1.13**	**-0.19**	**-0.21**	**-0.64**
XRP	–	–	**0.05**	**0.09**	**0.17**	**-0.24**
ZEC	-0.05	**1.00**	**0.33**	**-0.08**	**-0.07**	**1.31**
ZIL	–	–	**0.07**	–	**-0.02**	**0.20**

Table A.IV (Continued) Adaptive LASSO predictive regression results

	WAVES	XLM	XMR	XRP	ZEC	ZIL
ADA	**0.62**	**1.62**	**0.88**	–	**0.28**	**1.81**
BAT	**0.68**	-0.19	**-0.25**	-0.15	**-0.51**	**0.39**
BNB	**3.00**	**1.51**	**1.05**	**1.40**	**0.56**	**4.14**
BTC	**8.02**	**1.84**	**10.93**	**1.48**	**5.48**	**8.48**
BTT	-0.08	**-0.39**	**-0.37**	–	**-0.12**	0.03
CELR	–	-0.12	**-0.17**	-0.09	**-0.07**	-0.06
EOS	**1.55**	**1.17**	**1.17**	**1.14**	**1.99**	**0.84**
ETC	**0.66**	–	**0.95**	**-0.32**	**1.58**	**0.59**
ETH	**3.08**	**2.33**	**2.44**	**0.52**	**3.08**	**1.58**
FET	**0.26**	–	**-0.30**	–	**-0.08**	**0.57**
HOT	**-0.17**	**-0.33**	**-0.22**	**-0.19**	**-0.46**	0.00
ICX	**0.55**	-0.15	**-0.02**	-0.14	**-0.09**	**0.90**
IOST	–	–	**-0.43**	**-0.26**	**-0.30**	**0.24**
IOTA	**-0.18**	**-0.45**	**-0.48**	**-0.57**	**-0.49**	**0.38**
LINK	**1.34**	**0.39**	**0.50**	**0.50**	**0.59**	**1.42**
LTC	**0.42**	**1.53**	**1.16**	**0.49**	**1.84**	**1.90**
NEO	**0.55**	**0.71**	**1.22**	–	**1.53**	**1.19**
ONT	**1.46**	–	**0.55**	**0.58**	**1.26**	**1.54**
PAX	**-1.46**	-0.50	**-1.57**	**-2.24**	**0.69**	**-3.38**
QTUM	**0.40**	-0.21	**0.56**	–	**0.61**	**0.38**
TRX	**1.95**	**1.47**	**1.77**	**0.69**	**2.01**	**1.79**
TUSD	**8.63**	–	**2.54**	**0.66**	**9.09**	–
USDC	–	–	–	**-0.94**	–	**-3.64**
VET	**0.46**	-0.11	**-0.30**	–	**-0.26**	**0.36**
WAVES	**-14.07**	–	**-0.11**	–	**-0.19**	**0.71**
XLM	**1.16**	**-7.68**	**0.65**	**2.12**	**0.97**	**1.84**
XMR	**-0.64**	**-0.67**	**-13.34**	**-0.51**	–	0.05
XRP	–	**3.87**	**-0.01**	**-3.44**	–	–
ZEC	**1.01**	**0.92**	**2.17**	**0.51**	**-7.44**	**1.26**
ZIL	**0.16**	–	**-0.26**	–	**-0.30**	**-13.98**

Table A.V Correlations among cryptocurrencies

Table A.V reports the correlations among the 30 cryptocurrencies in the sample. The sample period is from 25 March 2019 to 30 April 2021.

	ADA	BAT	BNB	BTC	BTT	CELR	EOS
ADA	1.00	0.38	0.50	0.60	0.31	0.16	0.56
BAT	0.38	1.00	0.37	0.41	0.24	0.13	0.38
BNB	0.50	0.37	1.00	0.58	0.32	0.17	0.52
BTC	0.60	0.41	0.58	1.00	0.33	0.18	0.65
BTT	0.31	0.24	0.32	0.33	1.00	0.12	0.32
CELR	0.16	0.13	0.17	0.18	0.12	1.00	0.16
EOS	0.56	0.38	0.52	0.65	0.32	0.16	1.00
ETC	0.50	0.35	0.46	0.56	0.29	0.15	0.59
ETH	0.62	0.42	0.58	0.80	0.33	0.17	0.69
FET	0.23	0.20	0.24	0.28	0.16	0.12	0.23
HOT	0.18	0.15	0.19	0.19	0.14	0.09	0.19
ICX	0.36	0.29	0.34	0.39	0.23	0.13	0.35
IOST	0.40	0.31	0.37	0.42	0.25	0.14	0.39
IOTA	0.44	0.35	0.41	0.47	0.27	0.15	0.44
LINK	0.43	0.31	0.40	0.49	0.24	0.13	0.44
LTC	0.57	0.38	0.53	0.71	0.31	0.15	0.68
NEO	0.52	0.38	0.48	0.57	0.31	0.16	0.55
ONT	0.51	0.39	0.47	0.53	0.32	0.16	0.54
PAX	0.07	0.03	0.07	0.11	0.03	0.01	0.09
QTUM	0.47	0.36	0.43	0.50	0.29	0.15	0.49
TRX	0.51	0.35	0.47	0.57	0.38	0.15	0.58
TUSD	0.10	0.04	0.10	0.16	0.04	0.01	0.12
USDC	0.09	0.04	0.09	0.14	0.03	0.01	0.11
VET	0.44	0.33	0.41	0.47	0.27	0.15	0.43
WAVES	0.34	0.27	0.31	0.37	0.21	0.12	0.33
XLM	0.55	0.37	0.45	0.55	0.28	0.14	0.52
XMR	0.45	0.34	0.43	0.54	0.28	0.15	0.47
XRP	0.50	0.33	0.44	0.53	0.28	0.14	0.56
ZEC	0.47	0.35	0.44	0.52	0.28	0.15	0.50
ZIL	0.34	0.27	0.31	0.35	0.21	0.12	0.33

Table A.V (Continued) Correlations among cryptocurrencies

	ETC	ETH	FET	HOT	ICX	IOST	IOTA	LINK
ADA	0.50	0.62	0.23	0.18	0.36	0.40	0.44	0.43
BAT	0.35	0.42	0.20	0.15	0.29	0.31	0.35	0.31
BNB	0.46	0.58	0.24	0.19	0.34	0.37	0.41	0.40
BTC	0.56	0.80	0.28	0.19	0.39	0.42	0.47	0.49
BTT	0.29	0.33	0.16	0.14	0.23	0.25	0.27	0.24
CELR	0.15	0.17	0.12	0.09	0.13	0.14	0.15	0.13
EOS	0.59	0.69	0.23	0.19	0.35	0.39	0.44	0.44
ETC	1.00	0.60	0.21	0.17	0.32	0.37	0.40	0.40
ETH	0.60	1.00	0.26	0.21	0.40	0.43	0.48	0.51
FET	0.21	0.26	1.00	0.11	0.19	0.20	0.22	0.19
HOT	0.17	0.21	0.11	1.00	0.14	0.16	0.17	0.14
ICX	0.32	0.40	0.19	0.14	1.00	0.29	0.32	0.28
IOST	0.37	0.43	0.20	0.16	0.29	1.00	0.35	0.31
IOTA	0.40	0.48	0.22	0.17	0.32	0.35	1.00	0.34
LINK	0.40	0.51	0.19	0.14	0.28	0.31	0.34	1.00
LTC	0.58	0.74	0.23	0.18	0.36	0.40	0.44	0.45
NEO	0.49	0.59	0.23	0.18	0.35	0.40	0.44	0.40
ONT	0.49	0.56	0.23	0.19	0.35	0.41	0.43	0.40
PAX	0.07	0.10	0.02	0.01	0.03	0.03	0.04	0.06
QTUM	0.45	0.52	0.21	0.17	0.33	0.38	0.40	0.37
TRX	0.50	0.59	0.22	0.17	0.33	0.37	0.41	0.40
TUSD	0.09	0.14	0.03	0.01	0.04	0.04	0.05	0.08
USDC	0.08	0.13	0.02	0.01	0.04	0.03	0.05	0.07
VET	0.40	0.48	0.21	0.16	0.31	0.35	0.37	0.35
WAVES	0.30	0.37	0.18	0.13	0.25	0.27	0.30	0.27
XLM	0.47	0.56	0.22	0.16	0.33	0.37	0.41	0.40
XMR	0.43	0.52	0.22	0.17	0.32	0.36	0.39	0.37
XRP	0.48	0.56	0.20	0.16	0.31	0.34	0.39	0.38
ZEC	0.46	0.53	0.21	0.16	0.32	0.37	0.39	0.38
ZIL	0.31	0.36	0.17	0.14	0.26	0.28	0.30	0.27

Table A.V (Continued) Correlations among cryptocurrencies

	LTC	NEO	ONT	PAX	QTUM	TRX	TUSD	USDC
ADA	0.57	0.52	0.51	0.07	0.47	0.51	0.10	0.09
BAT	0.38	0.38	0.39	0.03	0.36	0.35	0.04	0.04
BNB	0.53	0.48	0.47	0.07	0.43	0.47	0.10	0.09
BTC	0.71	0.57	0.53	0.11	0.50	0.57	0.16	0.14
BTT	0.31	0.31	0.32	0.03	0.29	0.38	0.04	0.03
CELR	0.15	0.16	0.16	0.01	0.15	0.15	0.01	0.01
EOS	0.68	0.55	0.54	0.09	0.49	0.58	0.12	0.11
ETC	0.58	0.49	0.49	0.07	0.45	0.50	0.09	0.08
ETH	0.74	0.59	0.56	0.10	0.52	0.59	0.14	0.13
FET	0.23	0.23	0.23	0.02	0.21	0.22	0.03	0.02
HOT	0.18	0.18	0.19	0.01	0.17	0.17	0.01	0.01
ICX	0.36	0.35	0.35	0.03	0.33	0.33	0.04	0.04
IOST	0.40	0.40	0.41	0.03	0.38	0.37	0.04	0.03
IOTA	0.44	0.44	0.43	0.04	0.40	0.41	0.05	0.05
LINK	0.45	0.40	0.40	0.06	0.37	0.40	0.08	0.07
LTC	1.00	0.55	0.53	0.10	0.48	0.56	0.14	0.12
NEO	0.55	1.00	0.54	0.06	0.50	0.50	0.09	0.09
ONT	0.53	0.54	1.00	0.06	0.50	0.49	0.08	0.06
PAX	0.10	0.06	0.06	1.00	0.04	0.08	0.19	0.20
QTUM	0.48	0.50	0.50	0.04	1.00	0.45	0.06	0.05
TRX	0.56	0.50	0.49	0.08	0.45	1.00	0.11	0.09
TUSD	0.14	0.09	0.08	0.19	0.06	0.11	1.00	0.24
USDC	0.12	0.09	0.06	0.20	0.05	0.09	0.24	1.00
VET	0.44	0.43	0.43	0.04	0.40	0.40	0.05	0.04
WAVES	0.34	0.33	0.33	0.02	0.31	0.30	0.04	0.03
XLM	0.52	0.48	0.47	0.06	0.45	0.47	0.08	0.07
XMR	0.48	0.45	0.43	0.05	0.41	0.42	0.06	0.05
XRP	0.55	0.46	0.45	0.07	0.42	0.49	0.10	0.09
ZEC	0.51	0.47	0.47	0.05	0.43	0.44	0.07	0.05
ZIL	0.34	0.33	0.34	0.02	0.31	0.30	0.02	0.02

Table A.V (Continued) Correlations among cryptocurrencies

	VET	WAVES	XLM	XMR	XRP	ZEC	ZIL
ADA	0.44	0.34	0.55	0.45	0.50	0.47	0.34
BAT	0.33	0.27	0.37	0.34	0.33	0.35	0.27
BNB	0.41	0.31	0.45	0.43	0.44	0.44	0.31
BTC	0.47	0.37	0.55	0.54	0.53	0.52	0.35
BTT	0.27	0.21	0.28	0.28	0.28	0.28	0.21
CELR	0.15	0.12	0.14	0.15	0.14	0.15	0.12
EOS	0.43	0.33	0.52	0.47	0.56	0.50	0.33
ETC	0.40	0.30	0.47	0.43	0.48	0.46	0.31
ETH	0.48	0.37	0.56	0.52	0.56	0.53	0.36
FET	0.21	0.18	0.22	0.22	0.20	0.21	0.17
HOT	0.16	0.13	0.16	0.17	0.16	0.16	0.14
ICX	0.31	0.25	0.33	0.32	0.31	0.32	0.26
IOST	0.35	0.27	0.37	0.36	0.34	0.37	0.28
IOTA	0.37	0.30	0.41	0.39	0.39	0.39	0.30
LINK	0.35	0.27	0.40	0.37	0.38	0.38	0.27
LTC	0.44	0.34	0.52	0.48	0.55	0.51	0.34
NEO	0.43	0.33	0.48	0.45	0.46	0.47	0.33
ONT	0.43	0.33	0.47	0.43	0.45	0.47	0.34
PAX	0.04	0.02	0.06	0.05	0.07	0.05	0.02
QTUM	0.40	0.31	0.45	0.41	0.42	0.43	0.31
TRX	0.40	0.3	0.47	0.42	0.49	0.44	0.30
TUSD	0.05	0.04	0.08	0.06	0.10	0.07	0.02
USDC	0.04	0.03	0.07	0.05	0.09	0.05	0.02
VET	1.00	0.29	0.41	0.38	0.38	0.39	0.31
WAVES	0.29	1.00	0.31	0.30	0.30	0.31	0.25
XLM	0.41	0.31	1.00	0.43	0.56	0.44	0.32
XMR	0.38	0.30	0.43	1.00	0.40	0.44	0.30
XRP	0.38	0.30	0.56	0.40	1.00	0.42	0.30
ZEC	0.39	0.31	0.44	0.44	0.42	1.00	0.31
ZIL	0.31	0.25	0.32	0.30	0.30	0.31	1.00

Table A.VI LASSO predictive regression results

Table A.VI reports coefficient estimates for each of the 30 cryptocurrencies from the LASSO predictive model. Return of each cryptocurrency in the top header serves as the dependent variable in the LASSO model. Lagged return of each cryptocurrency in the left column serves as the independent variable. − indicates that the corresponding cryptocurrency in the left column was not selected as a return predictor by the LASSO. Estimates are presented in bps and those in bold are statistically significant based on their bootstrapped 90% confidence intervals. The sample period is from 25 March 2019 to 30 April 2021.

	ADA	BAT	BNB	BTC	BTT	CELR
ADA	-0.08	**0.00**	**0.02**	**0.02**	**0.00**	**0.00**
BAT	**0.02**	-0.14	**0.04**	**0.01**	**0.00**	**0.00**
BNB	**0.00**	**0.00**	-0.06	**0.03**	−	−
BTC	−	−	−	−	−	−
BTT	**0.00**	**0.00**	**0.01**	**0.05**	-0.08	−
CELR	**0.01**	**0.01**	**0.05**	**0.08**	**0.01**	-0.12
EOS	−	−	**0.01**	**0.01**	−	−
ETC	**0.00**	**0.00**	**0.01**	**0.04**	**0.00**	**0.00**
ETH	**0.00**	**0.00**	**0.00**	**0.03**	**0.00**	**0.00**
FET	**0.02**	**0.00**	**0.05**	**0.06**	**0.01**	**0.02**
HOT	**0.01**	**0.00**	**0.06**	**0.08**	**0.02**	**0.01**
ICX	**0.02**	**0.01**	**0.04**	**0.03**	−	**0.00**
IOST	**0.01**	−	**0.02**	**0.10**	−	−
IOTA	**0.02**	**0.00**	**0.03**	**0.06**	**0.00**	**0.00**
LINK	**0.01**	−	**0.03**	**0.05**	**0.00**	**0.00**
LTC	**0.00**	−	**0.01**	**0.03**	−	−
NEO	**0.01**	**0.00**	**0.02**	**0.03**	−	**0.00**
ONT	**0.01**	**0.00**	**0.02**	**0.03**	**0.00**	−
PAX	−	−	**0.00**	**-0.01**	−	−
QTUM	**0.01**	**0.00**	**0.01**	**0.06**	−	**0.00**
TRX	**0.01**	**0.00**	**0.01**	**0.01**	**0.02**	−
TUSD	**0.00**	−	**0.00**	**-0.01**	**0.00**	**0.00**
USDC	−	**0.00**	**0.00**	**-0.01**	**0.00**	**0.00**
VET	**0.01**	**0.00**	**0.02**	**0.08**	−	−
WAVES	**0.01**	**0.01**	**0.03**	**0.08**	**0.00**	−
XLM	**0.01**	**0.00**	**0.01**	**0.01**	0.00	**0.00**
XMR	**0.01**	**0.00**	**0.01**	**0.11**	**0.00**	**0.00**
XRP	−	−	**0.01**	**0.01**	−	−
ZEC	**0.00**	**0.00**	**0.00**	**0.06**	−	**0.00**
ZIL	**0.02**	**0.00**	**0.04**	**0.09**	**0.00**	−

Table A.VII Summary of major market-wide events

Table A.VII lists the nine major events included in our sample, based on their broad impact on the cryptocurrency market.

No	Event	Date
1	PlusToken Ponzi Scheme	2019/06/29
2	Donald Trump lambasted cryptos on Twitter	2019/07/11
3	President Xi Jinping's advocacy on blockchain	2019/10/24
4	Black Thursday	2020/03/12
5	Elon Musk tweeted doge	2020/12/20
6	Rumor on Bitcoin double spending	2021/01/21
7	Elon Musk added a bitcoin tag on his Twitter bio	2021/01/29
8	PayPal launched "Checkout with Crypto"	2021/03/30
9	Coinbase's IPO	2021/04/14

Table A.VIII Long-short portfolio performance on event day and normal days

Table A.VIII reports the long-short portfolio alphas with different regression specifications on the event and normal days. We identify a total of nine major events and classify the remaining days in our sample as normal days. Panel A (B) reports the alphas on event days (normal days) and Panel C reports the differences between alphas on event days and normal days. Column (1) shows the excess returns. Column (2) – (6) present alphas controlling for one risk factor (MKT, SMB, MOM5, MOM10, or MOM30). Column (7) presents alphas controlling for all five factors in the regression. Newey-West adjusted t-statistics are shown in the parentheses. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Event days							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-2.67***	-2.31***	-2.52***	-3.28***	-3.34***	-3.39***	-2.27***
2	-0.99***	-0.60***	-0.82***	-1.81***	-1.79***	-1.81***	-0.71***
3	-0.44**	-0.08	-0.28*	-1.23**	-1.26***	-1.23***	-0.27*
4	0.03	0.42***	0.21	-0.95*	-0.97*	-0.88**	0.08
Long	1.80***	2.14***	1.95***	0.94**	0.99**	1.07***	1.87***
Long - Short	4.47***	4.45***	4.47***	4.22***	4.33***	4.46***	4.15***
T-stats	*(13.57)*	*(9.84)*	*(10.51)*	*(14.42)*	*(15.14)*	*(12.67)*	*(16.93)*
Panel B: Normal days							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.59***	-1.63***	-1.61***	-1.72***	-1.77***	-1.80***	-1.61***
2	-0.27***	-0.30***	-0.28***	-0.57***	-0.56***	-0.54***	-0.46***
3	0.02*	-0.01	0.01	-0.25***	-0.24***	-0.21***	-0.12***
4	0.33***	0.29***	0.31***	0.06	0.08**	0.13***	0.20***
Long	1.74***	1.70***	1.72***	1.43***	1.50***	1.62***	1.60***
Long - Short	3.33***	3.33***	3.33***	3.15***	3.27***	3.42***	3.21***
T-stats	*(72.15)*	*(86.13)*	*(81.73)*	*(79.71)*	*(88.74)*	*(91.15)*	*(81.51)*
Panel C: Difference between event days and normal days							
	Excess Return	MKT	SMB	MOM5	MOM10	MOM30	ALL
Short	-1.07**	-0.66**	-0.95**	-1.08**	-1.09**	-1.06**	-0.63**
2	-0.73**	-0.30	-0.59*	-0.74**	-0.75**	-0.71*	-0.30
3	-0.46**	-0.03	-0.32*	-0.48**	-0.48**	-0.44*	-0.03
4	-0.29*	0.14	-0.15	-0.31	-0.32	-0.28	0.14
Long	0.07	0.50**	0.19	0.05	0.05	0.08	0.49**
Long - Short	1.14*	1.15**	1.15**	1.13**	1.14**	1.14**	1.12**
T-stats	*(1.93)*	*(2.22)*	*(2.12)*	*(2.37)*	*(2.37)*	*(2.17)*	*(2.37)*

Table A.IX ADF test results

Table A.IX reports the Augmented Dickey-Fuller (ADF) test results. For each coin, ADF-stats and p-values are reported. The sample period is from 25 March 2019 to 30 April 2021. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

Coin	ADF-stat	p-value	Coin	ADF-stat	p-value
ADA	-101.48	0.01	LTC	-99.05	0.01
BAT	-99.99	0.01	NEO	-99.67	0.01
BNB	-101.76	0.01	ONT	-100.25	0.01
BTC	-98.09	0.01	PAX	-109.67	0.01
BTT	-99.74	0.01	QTUM	-98.57	0.01
CELR	-98.57	0.01	TRX	-99.24	0.01
EOS	-99.15	0.01	TUSD	-113.28	0.01
ETC	-100.58	0.01	USDC	-110.41	0.01
ETH	-98.97	0.01	VET	-100.75	0.01
FET	-99.91	0.01	WAVES	-99.28	0.01
HOT	-101.30	0.01	XLM	-100.62	0.01
ICX	-98.43	0.01	XMR	-97.12	0.01
IOST	-97.60	0.01	XRP	-99.96	0.01
IOTA	-99.26	0.01	ZEC	-99.98	0.01
LINK	-99.97	0.01	ZIL	-99.59	0.01

Table A.X Portfolio performance using 5-minute rebalancing frequency

Table A.X reports portfolio returns (in bps) with and without trading costs in the spot and futures markets. For each five minutes, we sort the 30 sample coins in ascending order based on the 1-minute excess return forecasts generated from the adaptive LASSO technique using half-day and one-day rolling windows and group them into five equal-weighted portfolios with a five-minute rebalancing frequency. We subsequently construct a long-short portfolio that goes long (short) the top (bottom) quintile portfolio. For the spot market, we follow Jia, Wu, Yan, and Liu (2023) to use 12 bps as the total transaction cost, including bid-ask spread and trading fees. For the futures market, we use 4 bps trading fees according to *Binance*. The sample period for spots is 25 March 2019 to 30 April 2021 and for futures is 29 July 2020 to 30 April 2021.The t-statistics are Newey-West adjusted. *, **, and *** represent statistical significance at the 10%, 5%, and 1% levels, respectively.

	Futures		**Spot**	
	Half-day Window	One-day Window	Half-day Window	One-day Window
Short	-0.41***	-0.58***	-1.92***	-2.38***
2	0.23**	0.11	-0.20***	-0.53***
3	0.31***	0.29***	0.11*	0.16***
4	0.45***	0.56***	0.50***	0.74***
Long	1.30***	1.49***	2.44***	2.96***
Long - Short (without Transaction Costs)	1.71***	2.07***	4.36***	5.34***
Long - Short (with Transaction Costs)	0.66***	0.94***	0.80***	1.60***

Figure A.1 Dollar trading volume in the cryptocurrency market

Figure A.1 illustrates the dollar trading volume of the 30 sample cryptocurrencies and all other cryptocurrencies in the spot market, on the day that we started the project, i.e., 9 May 2020, based on data obtained from *Binance*. A total of 107 cryptocurrencies were traded on that day.

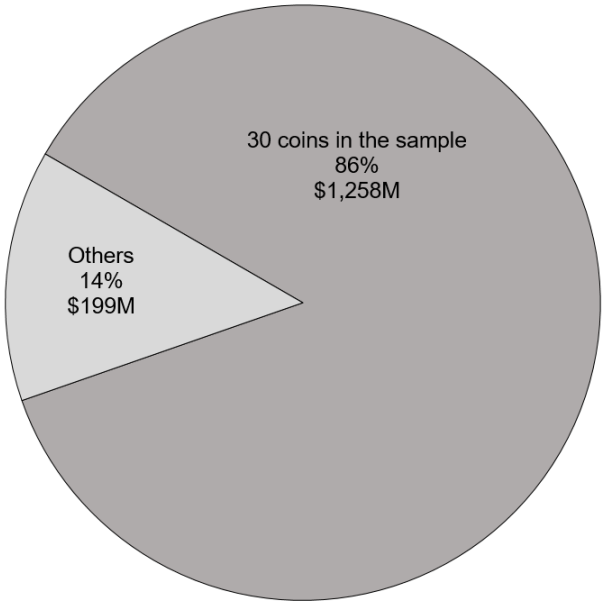
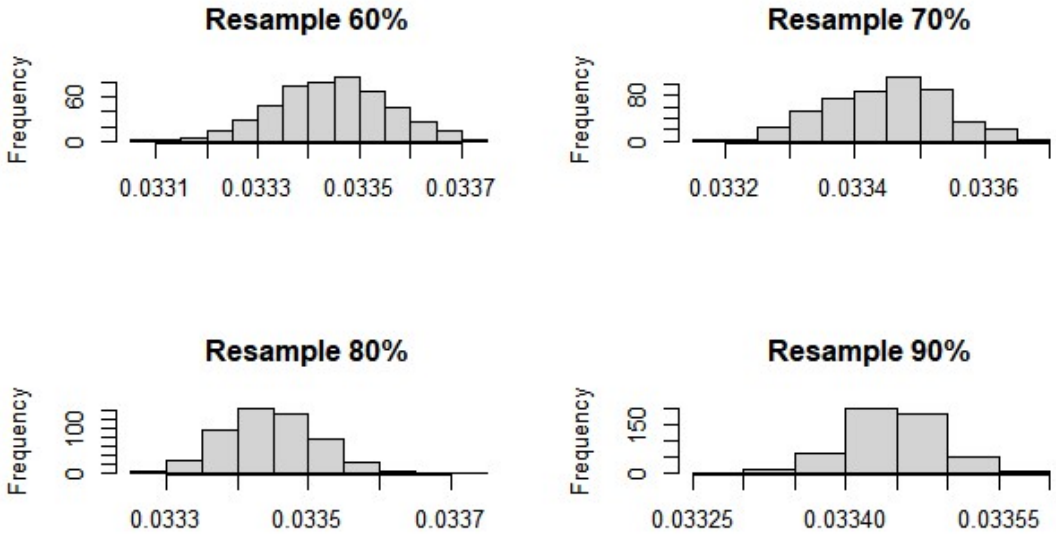


Figure A.2 Permutation test results

Figure A.2 plots the permutation/randomization test results for the sample periods (Panel A) and the number of coins (Panel B). The frequency distributions of the corresponding long-short portfolio returns are reported. In Panel A, we randomly select 60%, 70%, 80%, or 90% of the data in terms of the sample periods and repeat the process 500 times. In Panel B, we randomly select only 20 or 25 coins in our sample 100 times. The sample period is from 25 March 2019 to 30 April 2021.

Panel A: Sample periods randomization



Panel B: Number of coins randomization

